If we demand that $|C| = 1$, then

$$G_4(x, x') = \begin{cases} -\dfrac{i}{C^* - 1}, & x > x', \\ +\dfrac{i}{C - 1}, & x \leq x', \end{cases} \tag{7.12'}$$

from which we see that $G_4(x, x')$ is symmetric in the sense that

$$G(x, x') = G(x', x)^* .$$

This definition of symmetry is motivated by the definition of a Hermitian matrix as being one whose elements satisfy

$$a_{ij} = a_{ji}^* .$$

As we shall see, the relationship between symmetric functions like $G_4(x, x')$ and Hermitian matrices is a very close and fruitful one. For the present, we may conclude that even in the simple case of a single derivative, the choice of boundary condition can mean the difference between a symmetric and a non-symmetric Green's function [since $G_1(x, x')$ of Eq. (7.3′) is not symmetric].

The fact that certain boundary conditions lead to a Green's function which is symmetric while others do not is readily made plausible. The reader can easily show that on the space of square-integrable functions which satisfy $f(1) = Cf(0)$ with $C = e^{i\psi}$ the differential operator $p = -id/dx$ is symmetric in the sense that $(f, pg) = (pf, g)$. Here $(\ ,\ )$ denotes the inner product of Eq. (5.3) with $a = 0$ and $b = 1$. However, p will not be symmetric if we require only that the functions in the space take on a fixed value at $x = 0$. Similar comments apply to the operator d^2/dx^2, if one remembers that in finding a Green's function for d^2/dx^2 acting on the space of square-integrable functions which take on fixed values at $x = 0$ and $x = 1$ one can always choose the endpoint values to be zero. This follows simply from the fact that if we solve

$$\frac{d^2\eta}{dx^2} = f(x)$$

subject to $\eta(0) = 0$ and $\eta(1) = 0$, then the solution to

$$\frac{d^2y}{dx^2} = f(x)$$

with $y(0) = y_0$ and $y(1) = y_1$ is just $y(x) = y_0 + (y_1 - y_0)x + \eta(x)$. Because the Green's functions $G_3(x, x')$ and $G_4(x, x')$ are symmetric functions of x and x', it is a simple matter to see that the integral operators K_3 and K_4 which have $G_3(x, x')$ and $G_4(x, x')$, respectively, as kernels are symmetric operators in the same sense that p was symmetric, that is, $(f, K_3g) = (K_3f, g)$ and $(f, K_4g) = (K_4f, g)$. Thus, it is not surprising that there appears to be a close relationship between symmetric differential operators and symmetric Green's functions.

The reader may well wonder why we have bothered developing such an elaborate way of integrating simple differential equations. One very important

reason is the following. Suppose that, instead of Eq. (7.1), we had to solve

$$-i\frac{dy}{dx} = f(x, y) \ . \tag{7.13}$$

This is a true differential equation in the sense that it is not immediately integrable. By our previous work we have

$$y(x) = \alpha + \int_0^1 G(x, x')f[x', y(x')]\, dx' \ . \tag{7.14}$$

If the boundary condition is $y(0) = y_0$, then $\alpha = y_0$, and $G(x, x') = -i\theta(x - x')$; if the boundary condition is $y(1) = Cy(0)$, where $|C| = 1$, then $\alpha = 0$, and $G(x, x')$ is given by the symmetric function $G_4(x, x')$ of Eq. (7.12). Since the unknown function $y(x)$ now appears under the integral sign, we have transformed the *differential* equation into an *integral* equation. Similarly, if we have

$$\frac{d^2y}{dx^2} = f(x, y)$$

with $y(0) = y_0$, $y(1) = y_1$, we get as the corresponding integral equation

$$y(x) = y_0 + (y_1 - y_0)x + \int_0^1 G_3(x, x')f[x', y(x')]\, dx' \ ,$$

where $G_3(x, x')$ is the symmetric function given by Eq. (7.11′). For example, the well-known second-order linear differential equation

$$\frac{d^2y}{dx^2} + \lambda y = 0 \ , \qquad y(0) = 0 = y(1)$$

is transformed into the integral equation

$$y(x) = -\lambda \int_0^1 G_3(x, x')y(x')\, dx' \ .$$

There is one very interesting and tricky aspect of these equations which should be mentioned here. Crudely speaking, the differential operator id/dx is *unbounded* in the sense that it can act on functions of unit norm in Hilbert space and transform them into functions with arbitrarily large norm. For example, the norm of e^{inx} in the Hilbert space of square-integrable functions on $[0, 1]$ is 1, but the norm of

$$i\frac{d}{dx}[e^{inx}] = -ne^{inx}$$

is n, which can be as large as we like. As we shall see later, the theory of such unbounded operators is rather subtle.

However, the integral operator whose kernel is $G_4(x, x')$ is *bounded*, in the sense that it transforms any square-integrable function of unit norm into a function whose norm is less than some given number. As a general rule, let

$$f(x) = \int_0^1 K(x, y)\, g(y)\, dy \ .$$

If we assume that $g(y)$ is square-integrable and that $K(x, y)$ is square-integrable in y, for all x, then by Schwartz's inequality (Eq. 4.10),

$$|f(x)| \leq \left[\int_0^1 |K(x, y)|^2 \, dy\right]^{1/2} \left[\int_0^1 |g(y)|^2 \, dy\right]^{1/2}.$$

Thus, if g has unit norm, we get

$$|f(x)| \leq \left[\int_0^1 |K(x, y)|^2 \, dy\right]^{1/2}.$$

But this means that

$$\int_0^1 |f(x)|^2 \, dx \leq \int_0^1 \int_0^1 |K(x, y)|^2 \, dx \, dy .$$

If $K(x, y)$ is square integrable in the xy-plane, which is the case for all the Green's functions discussed above, then calling

$$\int_0^1 \int_0^1 |K(x, y)|^2 \, dx \, dy = C ,$$

we have

$$\int_0^1 |f(x)|^2 \, dx \equiv ||f||^2 \leq C$$

for *any* function $g(x)$ of unit norm. Kernels of the square-integrable type actually have many other nice properties. For instance, the kernel $G_4(x, x')$ transforms the sequence of functions of unit norm $\{e^{inx}\}$ into a sequence of functions whose norms tend toward zero. In fact, for kernels of this type, *any* infinite bounded sequence of functions (i.e., functions $f_n(x)$ satisfying $||f_n|| \leq C$ for all n) will be transformed into a sequence containing a subsequence which is convergent in the mean. Operators of this type are said to be *completely continuous*, and we shall have a great deal more to say about them in Chapter 9. The analysis of these operators is much more straightforward than that of unbounded operators, and, in fact, bears a very close resemblance to ordinary matrix theory. Thus by turning a differential equation into an integral equation, or, in other words, by turning a differential operator into an integral operator, it is often possible to effect significant mathematical simplifications in a given problem.

7.2 GREEN'S FUNCTIONS AND DELTA FUNCTIONS

With the general idea of a Green's function in mind, let us now approach the problem of solving a differential equation in a slightly different manner. Suppose that we have the equation

$$Ly = f , \tag{7.15}$$

where L is some linear ordinary differential operator and f is a given function. Throwing aside questions of rigor, suppose now that L possesses a complete,

orthonormal set of eigenfunctions $\{\phi_n(x)\}$ so that

$$L\phi_n(x) = \lambda_n\phi_n(x) .$$

Under these circumstances we could write (omitting the dots over the equal signs which denote mean convergence)

$$y(x) = \sum_{n=1}^{\infty} \alpha_n\phi_n(x) , \tag{7.16a}$$

$$f(x) = \sum_{n=1}^{\infty} \beta_n\phi_n(x) . \tag{7.16b}$$

The sum runs to infinity, which reflects the fact that in an infinite-dimensional Hilbert space, a complete orthonormal set contains an infinite number of elements.

If we now substitute Eqs. (7.16a) and (7.16b) into Eq. (7.15), we get

$$Ly = L\sum_{n=1}^{\infty} \alpha_n\phi_n = \sum_{n=1}^{\infty} L\alpha_n\phi_n = \sum_{n=1}^{\infty} \alpha_n\lambda_n\phi_n = f = \sum_{n=1}^{\infty} \beta_n\phi_n .$$

Thus

$$\sum_{n=1}^{\infty} (\alpha_n\lambda_n - \beta_n)\phi_n = 0 .$$

Since the ϕ_n are linearly independent, we conclude that

$$\alpha_n = \beta_n/\lambda_n;$$

thus

$$y(x) = \sum_{n=1}^{\infty} \frac{1}{\lambda_n} \beta_n\phi_n(x) , \tag{7.17}$$

where $\beta_n = (\phi_n, f)$. If some $\lambda_n = 0$, then a solution exists only if the corresponding $\beta_n = 0$. In this case, the solution will not be unique; an arbitrary multiple of the ϕ_n corresponding to $\lambda_n = 0$ can be added to any solution (see Theorem 3.17). For the moment, then, we assume that $\lambda_n \neq 0$ for all n.

Let us write Eq. (7.17) in more detail:

$$\begin{aligned} y(x) &= \sum_{n=1}^{\infty} \frac{1}{\lambda_n} \phi_n(x)(\phi_n, f) \\ &= \sum_{n=1}^{\infty} \frac{1}{\lambda_n} \phi_n(x) \int \phi_n^*(x')f(x')\, dx' \\ &= \int \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n^*(x')}{\lambda_n} f(x')\, dx' . \end{aligned}$$

Thus

$$y(x) = \int G(x, x')f(x')\, dx' , \tag{7.18}$$

where

$$G(x, x') = \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n^*(x')}{\lambda_n} . \tag{7.18'}$$

(Here we have interchanged sums and integrals in a rather cavalier manner, i.e., without establishing the uniform convergence of the sum.) Thus we obtain a representation of the Green's function in terms of an infinite eigenfunction expansion. If the λ_n are real, then $G(x, x')$ is symmetric:

$$G(x, x') = G(x', x)^* .$$

As an example, consider the operator

$$L = \frac{d^2}{dx^2}$$

for x in $[0, 1]$. The normalized eigenfunctions of L on $[0, 1]$, which vanish at the endpoints, are $\phi_n(x) = \sqrt{2} \sin (n\pi x)$, $n = 1, 2, \cdots$, with eigenvalues $\lambda_n = -n^2\pi^2$. Thus the Green's function for this case is

$$G(x, x') = -\frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{\sin (n\pi x) \sin (n\pi x')}{n^2} . \tag{7.19}$$

Since the series is dominated term by term by $1/n^2$, we have uniform convergence to a continuous function. In fact, the function to which the series of Eq. (7.19) converges must be precisely the function $G_3(x, x')$ given by Eq. (7.11′), since we have already determined by another (more rigorous) method that this is the Green's function of d^2/dx^2 with the two-endpoint boundary condition. It is interesting to calculate $G_3(x, x')$ at a few points to see whether Eqs. (7.11′) and (7.19) agree. For example, at $x = x' = \frac{1}{2}$, $G_3(\frac{1}{2}, \frac{1}{2}) = -\frac{1}{4}$, while Eq. (7.19) gives

$$G\left(\frac{1}{2}, \frac{1}{2}\right) = -\frac{2}{\pi^2} \sum_{n=1}^{\infty}{}' \frac{1}{n^2} ,$$

where the prime on the summation sign means that the sum runs only over odd values of n. Using the results of Eq. (6.61) and Problem (6.32) one readily obtains the result

$$\sum_{n=1}^{\infty}{}' \frac{1}{n^2} = \frac{\pi^2}{8} ,$$

so that $G(\frac{1}{2}, \frac{1}{2}) = -\frac{1}{4}$, in agreement with the previously obtained result for $G_3(\frac{1}{2}, \frac{1}{2})$.

It is also instructive to look at the operator $p = -id/dx$ acting on the space of square-integrable functions on $[0, 1]$ which satisfy $f(1) = Cf(0)$, where $C = e^{i\psi}$ $(0 \leq \psi < 2\pi)$. We have considered this operator from a different point of view in the previous section. The normalized eigenfunctions of p are readily seen to be $\phi_n = e^{i(2\pi n + \psi)x}$ $(n = 0, \pm 1, \pm 2, \cdots)$; the corresponding eigenvalues

are $\lambda_n = 2\pi n + \psi$. Thus, the Green's function for this operator is

$$G(x, x') = \sum_{n=-\infty}^{\infty} \frac{e^{i(2\pi n+\psi)(x-x')}}{2\pi n + \psi} .$$

Note that if $\psi = 0$, that is, if $C = 1$, this expression for $G(x, x')$ is ill-defined because of a zero denominator when $n = 0$. This illustrates the type of difficulty that arises when an operator has an eigenvector with a zero eigenvalue. In the above case, $\lambda_0 = 0$, so the solution of Eq. (7.17) breaks down, and the problem posed in Eq. (7.15) does not have a solution unless $(\phi_0, f) = 0$. Even if $(\phi_0, f) = 0$, the solution to Eq. (7.15) is not unique. However, if $\psi \neq 0$, the problem is well-posed, and it is, in fact, a simple matter to determine $G(x, x')$. We shall see in the next section that the Green's function for p must be constant except when $x = x'$. To evaluate $G(x, x')$ when $x > x'$ we may take $x - x' = 1/2$ without loss of generality. Thus,

$$G(x, x') = \frac{1}{2\pi} e^{i\psi/2} \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{n + \psi/2\pi} \qquad (x > x') ,$$

and similarly, taking $x - x' = -\frac{1}{2}$,

$$G(x, x') = \frac{1}{2\pi} e^{-i\psi/2} \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{n + \psi/2\pi} \qquad (x < x') .$$

From the result of Problem 6.32 (see also Eq. (6.57)) the sum in question can be readily evaluated; the reader will find that the Green's function obtained in this way agrees precisely with the Green's function $G_4(x, x')$ of Eq. (7.12′) which was obtained in a more elementary fashion for the operator p.

An interesting way of viewing the Green's function is to operate on $G(x, x')$ with L:

$$LG(x, x') = L \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n^*(x')}{\lambda_n} .$$

If we interchange summation and the operation denoted by L, we have, using $L\phi_n(x) = \lambda_n\phi_n(x)$,

$$LG(x, x') = \sum_{n=1}^{\infty} \frac{L\phi_n(x)\phi_n^*(x')}{\lambda_n} = \sum_{n=1}^{\infty} \phi_n(x)\phi_n^*(x') \equiv I(x, x') .$$

For any function $f(x)$,

$$\int I(x, x')f(x')\,dx' = \sum_{n=1}^{\infty} \phi_n(x) \int \phi_n^*(x')f(x')\,dx'$$

$$= \sum_{n=1}^{\infty} \phi_n(x)(\phi_n, f) .$$

But since the ϕ_n are assumed to be a complete orthonormal set, this equation may be written

$$\int I(x, x')f(x')\,dx = f(x)\,.$$

We have already seen in Chapter 5 that the "function" $I(x, x')$ which has this property is just the Dirac δ-function, $\delta(x - x')$. Thus we may write

$$LG(x, x') = \delta(x - x')\,, \tag{7.20}$$

an equation which is perhaps better written in the form

$$LK = I\,, \tag{7.21}$$

where K is the integral operator defined by

$$Kf = \int G(x, x')f(x')\,dx'\,.$$

Thus, in terms of Eq. (7.21), finding the Green's function of an operator L is tantamount to *inverting* L, so that equations like $Ly = f$ can be solved according to

$$y = L^{-1}f = Kf = \int G(x, x')f(x')\,dx'\,.$$

This analysis tells us that if we go back to our simplest of Green's functions, $G_1(x, x')$ of Eq. (7.3′), we should find that

$$-i\frac{d}{dx}G_1(x, x') = \delta(x - x')\,.$$

Using Eq. (7.3′), we find that this becomes

$$\frac{d}{dx}\theta(x - x') = \delta(x - x')$$

or

$$\frac{d}{dt}\theta(t) = \delta(t)\,. \tag{7.22}$$

Let us see whether $d\theta/dt$ has the properties of a δ-function discussed in Section 5.3. Using the definition of $\theta(t)$, we see that

$$\int_{-\epsilon}^{\epsilon}\frac{d\theta(t)}{dt}\,dt = \theta(\epsilon) - \theta(-\epsilon) = 1\,,$$

so one basic property of the δ-function is trivially satisfied. Also, since $\theta(t)$ is a constant everywhere except at $t = 0$, $d\theta/dt$ vanishes everywhere except at $t = 0$. Now we investigate the effect of $d\theta(t)/dt$ as a factor in an integrand:

$$\int_{-\epsilon}^{\epsilon}\frac{d\theta(t)}{dt}f(t)\,dt = [\theta(t)f(t)\,]_{-\epsilon}^{\epsilon} - \int_{-\epsilon}^{\epsilon}\theta(t)f'(t)\,dt\,,$$

where we have integrated once by parts. Using the definition of $\theta(t)$, we get

$$\int_{-\epsilon}^{\epsilon} \frac{d\theta}{dt} f(t)\, dt = f(\epsilon) - \int_0^{\epsilon} f'(t)\, dt$$
$$= f(\epsilon) - [f(\epsilon) - f(0)] = f(0) .$$

Hence $d\theta/dt$ has the most important of the δ-function properties, namely,

$$\int_{-\epsilon}^{\epsilon} \delta(t) f(t)\, dt = f(0) .$$

An intuitive way of seeing that $d\theta/dt = \delta(t)$ is to write an approximate $\theta(t)$ as

$$\theta_\epsilon(t) = \begin{cases} 0 , & t < -\epsilon , \\ \frac{1}{2} + \frac{1}{2\epsilon} t , & -\epsilon \leq t \leq \epsilon , \\ 1 , & t > \epsilon . \end{cases}$$

Clearly, $\theta_\epsilon(t) \to \theta(t)$ as $\epsilon \to 0$. Now we compute $d\theta_\epsilon(t)/dt$ and find that

$$\theta'_\epsilon(t) = \begin{cases} 0 , & t < -\epsilon , \\ 1/2\epsilon , & -\epsilon \leq t \leq \epsilon , \\ 0 , & t > \epsilon . \end{cases}$$

The sequence of functions $\{\theta'_\epsilon(t)\}$ is one of those we showed in Section 5.3, which approximated the δ-function. Figure 7.3 shows the function $\theta_\epsilon(t)$ and $\theta'_\epsilon(t)$. With these properties of the δ-function in mind, the reader should verify that all the Green's functions derived in the last section satisfy Eq. (7.20).

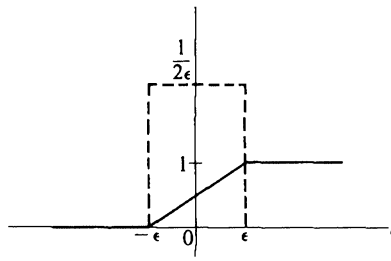

Fig. 7.3 The solid line represents the function $\theta_\epsilon(t)$; the dashed line represents the function $\theta'_\epsilon(t)$.

The differential representation of the δ-function in Eq. (7.22) is very useful, and for future reference we note a few of its simple consequences. First, since

$$\theta(at) = \theta(t) \quad \text{if} \quad a > 0 ,$$
$$\theta(at) = 1 - \theta(t) \quad \text{if} \quad a < 0 ,$$

we obtain by differentiation

$$a\frac{d\theta(at)}{d(at)} = \frac{d\theta}{dt}, \qquad a > 0,$$
$$a\frac{d\theta(at)}{d(at)} = -\frac{d\theta}{dt}, \qquad a < 0.$$

Thus

$$a\delta(at) = \delta(t), \qquad a > 0,$$
$$a\delta(at) = -\delta(t), \qquad a < 0,$$

or

$$\delta(at) = \frac{1}{|a|}\delta(t). \tag{7.23}$$

A second useful result concerns the function signum(x), which is usually written sgn (x). This is a function which equals $+1$ when x is positive and -1 when x is negative. Clearly,

$$\text{sgn}\,(x) = \theta(x) - \theta(-x),$$

so

$$\frac{d}{dx}[\text{sgn}\,(x)] = \frac{d\theta(x)}{dx} + \frac{d\theta(-x)}{d(-x)} = \delta(x) + \delta(-x) = 2\delta(x),$$

that is,

$$\frac{d}{dx}\text{sgn}\,(x) = 2\delta(x). \tag{7.24}$$

As an immediate consequence, we see that

$$\frac{d^2}{dx^2}|x| = \frac{d^2}{dx^2}[x\,\text{sgn}\,(x)] = \frac{d}{dx}[\text{sgn}\,(x) + 2x\delta(x)] = \frac{d}{dx}[\text{sgn}\,(x)] = 2\delta(x).$$

Here we have used the fact that $x\delta(x) = 0$ and Eq. (7.24).

With these rather informal interpretations of Green's functions and the δ-function in mind, we turn to some applications.

7.3 GREEN'S FUNCTIONS IN ONE DIMENSION

Having studied a few simple operators by way of introduction, we shall now try our hand at operators of a more general type and at the same time introduce another technique for the study of Green's functions. We shall begin with an example from classical mechanics, but before we are finished, it will be apparent that the applications arising from this special case are considerable indeed.

We consider the equation of the forced, damped harmonic oscillator,

$$\ddot{x}(t) + 2\gamma\dot{x}(t) + \omega_0^2 x(t) = F(t), \tag{7.25}$$

where we indicate differentiation with respect to time by a dot. We have chosen units of mass such that $m = 1$; γ is essentially the coefficient of frictional drag, ω_0 is the square root of the (positive) spring constant, and $F(t)$ is the external driving force. We assume that $F(t)$ has a Fourier transform,

$$\hat{F}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikt} F(t) dt .$$

Since we have assumed that $F(t)$ has a Fourier transform, $F(t)$ must go to zero for large $|t|$. This, coupled with the presence of damping, makes it reasonable to assume that $x(t)$ and its first derivative go to zero for large $|t|$, and therefore their Fourier transforms exist. Taking the Fourier transform of both sides of Eq. (7.25), we get

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x) e^{ikt}\, dt = \hat{F}(k) .$$

Let

$$\hat{x}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikt} x(t)\, dt;$$

then

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d^2x}{dt^2} e^{ikt}\, dt + 2\gamma \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dx}{dt} e^{ikt}\, dt + \omega_0^2 \hat{x}(k) = \hat{F}(k) . \tag{7.26}$$

In the first term we integrate by parts once to get

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{d^2x}{dt^2} e^{ikt} dt &= \left[\frac{dx}{dt} e^{ikt}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(\frac{dx}{dt}\right) \frac{d}{dt} (e^{ikt})\, dt \\ &= -ik \int_{-\infty}^{\infty} \left(\frac{dx}{dt}\right) e^{ikt}\, dt . \end{aligned}$$

Here the contribution from the endpoints vanishes since we assume that $x(t)$ and $\dot{x}(t)$ may be Fourier transformed and therefore vanish at large distances. Thus, Eq. (7.26) implies that

$$\frac{(2\gamma - ik)}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{dx}{dt}\right) e^{ikt}\, dt + \omega_0^2 \hat{x}(k) = \hat{F}(k) . \tag{7.27}$$

Finally, we integrate by parts once more and use the boundary conditions at infinity to find

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{dx}{dt}\right) e^{ikt} = -ik \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x(t) e^{ikt}\, dt = -ik\hat{x}(k) .$$

Hence Eq. (7.27) becomes

$$(-k^2 - 2i\gamma k + \omega_0^2)\hat{x}(k) = \hat{F}(k) . \tag{7.28}$$

Since $(\omega_0^2 - 2i\gamma k - k^2)$ has no zeroes on the real line,

$$\hat{x}(k) = \frac{\hat{F}(k)}{-k^2 - 2i\gamma k + \omega_0^2} ,$$

and by the basic result on Fourier transforms,

$$x(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\hat{F}(k)}{-k^2 - 2i\gamma k + \omega_0^2} e^{-ikt}\, dk .$$

Using the definition of $\hat{F}(k)$, we get

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt' \int_{-\infty}^{\infty} dk \frac{F(t')e^{ikt'}}{-k^2 - 2i\gamma k + \omega_0^2} e^{-ikt} .$$

Finally, we can add an arbitrary linear combination of solutions to the homogeneous equation to get the general solution

$$x(t) = Ax_1(t) + Bx_2(t) + \int_{-\infty}^{\infty} G(t, t')F(t')\, dt' , \tag{7.29a}$$

where

$$G(t, t') = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-ik(t-t')}}{-k^2 - 2i\gamma k + \omega_0^2} dk , \tag{7.29b}$$

and A and B must be chosen to satisfy the boundary conditions. For the two linearly independent solutions of the homogeneous equation, we may take

$$\begin{aligned} x_1(t) &= e^{-\gamma t} \sin [\sqrt{\omega_0^2 - \gamma^2}\, t] , \\ x_2(t) &= e^{-\gamma t} \cos [\sqrt{\omega_0^2 - \gamma^2}\, t] . \end{aligned} \tag{7.30}$$

Let us now see if we can evaluate $G(t, t')$. First we write

$$G(t, t') = -\frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-ik(t-t')}\, dk}{(k - k_1)(k - k_2)} , \tag{7.31}$$

where $k_1 = (\omega_0^2 - \gamma^2)^{1/2} - i\gamma$ and $k_2 = -(\omega_0^2 - \gamma^2)^{1/2} - i\gamma$. We will assume for definiteness that $\omega_0 > \gamma$, but this is in no way essential in what follows. The important fact about k_1 and k_2 is that their imaginary parts are always *negative*. Our work on complex variable theory puts us in a position to do this integral easily: we consider the integral in Eq. (7.31) as part of a contour integral in the complex k-plane. The relevant contour—which will be labeled C—runs along the real axis from $-K$ to $+K$ and then closes in the lower half-plane by

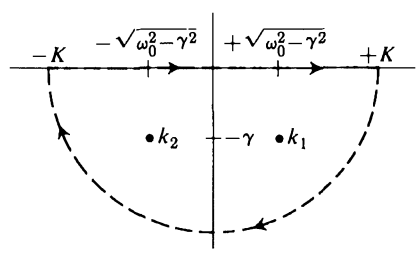

Fig. 7.4 The complex k-plane showing the contour used in evaluating the Green's function for the damped harmonic oscillator.

means of a semicircle of radius K about the origin (Fig. 7.4). We shall see that when $t - t' > 0$, this is the appropriate contour, whereas when $t - t' < 0$, the appropriate contour is one that closes in the upper half-plane:

$$I \equiv -\frac{1}{2\pi}\oint_C \frac{e^{-ik(t-t')}\,dk}{(k-k_1)(k-k_2)} = -\frac{1}{2\pi}\int_{-K}^{K} \frac{e^{-ik(t-t')}}{(k-k_1)(k-k_2)}\,dk$$
$$-\frac{1}{2\pi}\int_0^{-\pi} \frac{e^{-iK(t-t')e^{i\phi}}}{(Ke^{i\phi}-k_1)(Ke^{i\phi}-k_2)}\,iKe^{i\phi}\,d\phi\,. \tag{7.32}$$

In the first term on the right of Eq. (7.32), the integral is taken along the real axis; in the second term we integrate along the semicircle of radius K, on which

$$k = Ke^{i\phi}\,, \qquad dk = iKe^{i\phi}\,d\phi\,,$$

and ϕ ranges from $\phi = 0$ to $\phi = -\pi$. Now we let K, the radius of the contour, become very large. Then since the imaginary parts of k_1 and k_2 are both in the lower half-plane, the contour C eventually encloses both of them, and we can evaluate the integral by the residue theorem. We get

$$-2\pi I = \lim_{K\to\infty}\oint_C \frac{e^{-ik(t-t')}}{(k-k_1)(k-k_2)}\,dk = (-1)(2\pi i)\left[\frac{e^{-ik_1(t-t')}}{k_1-k_2} + \frac{e^{-ik_2(t-t')}}{k_2-k_1}\right].$$

The factor (-1) enters because the contour is closed in the *lower* half-plane, and so is traversed *clockwise*. Inserting the values of k_1 and k_2, we obtain

$$I = \frac{e^{-\gamma(t-t')}\sin\left[\sqrt{\omega_0^2-\gamma^2}(t-t')\right]}{\sqrt{\omega_0^2-\gamma^2}}\,.$$

As $K \to \infty$, the first term on the right-hand side of Eq. (7.32) becomes just $G(t, t')$, Eq. (7.31), so

$$\frac{e^{-\gamma(t-t')}\sin\left[\sqrt{\omega_0^2-\gamma^2}(t-t')\right]}{\sqrt{\omega_0^2-\gamma^2}} = G(t, t') + \lim_{K\to\infty} R\,, \tag{7.33}$$

where

$$R = -\frac{1}{2\pi}\int_0^{-\pi} \frac{e^{-iKe^{i\phi}(t-t')}}{(Ke^{i\phi}-k_1)(Ke^{i\phi}-k_2)}\,iKe^{i\phi}\,d\phi\,.$$

But this integral vanishes as $K \to \infty$ by Jordan's lemma (Section 6.8). Therefore

$$G(t, t') = \frac{e^{-\gamma(t-t')}\sin\left[\sqrt{\omega_0^2-\gamma^2}(t-t')\right]}{\sqrt{\omega_0^2-\gamma^2}} \qquad t > t'\,. \tag{7.34}$$

Now if $t < t'$, the above argument fails because the integrals involved in estimating R grow exponentially. In this case, however, we choose a contour that closes in the upper half-plane. But then the integrand in I of Eq. (7.32) has no singularities inside the contour, so

$$G(t, t') = 0\,, \qquad t < t'\,.$$

Thus we may summarize by saying

$$G(t, t') = \theta(t - t') \frac{e^{-\gamma(t-t')} \sin [\sqrt{\omega_0^2 - \gamma^2}\,(t - t')]}{\sqrt{\omega_0^2 - \gamma^2}}, \tag{7.35a}$$

and writing the general solution as

$$x(t) = Ax_1(t) + Bx_2(t) + \int_{t_0}^{t} \frac{e^{-\gamma(t-t')} \sin [\sqrt{\omega_0^2 - \gamma^2}(t - t')]\, F(t')\, dt'}{\sqrt{\omega_0^2 - \gamma^2}}, \tag{7.35b}$$

where t_0 is the time at which the initial conditions are applied (i.e., $F(t')$ may be taken to be zero at times prior to t_0). Note that the term involving the integral vanishes at $t = t_0$, as does its first derivative with respect to t. Thus A and B are determined from the initial conditions, $x(t_0) = x_0$, $\dot{x}(t_0) = \dot{x}_0$, just as they would be if there were no inhomogeneous term present. We anticipated this result in referring to the kernel of the integral as $G(t, t')$ at the beginning of this section.

Finally, we leave it to the reader to show that

$$\left[\frac{d^2}{dt^2} + 2\gamma \frac{d}{dt} + \omega_0^2\right] G(t, t') = \delta(t - t'), \tag{7.36}$$

just as we expect.

As an example, suppose that the forcing term is

$$F(t) = F_0 e^{-\alpha t}$$

starting at $t = 0$. The system is assumed to be at rest in its equilibrium position at $t = 0$, so that $A = 0 = B$ in Eq. (7.35b). Thus

$$x(t) = F_0 e^{-\gamma t} \int_0^t \frac{1}{\sqrt{\omega_0^2 - \gamma^2}} \sin [\sqrt{\omega_0^2 - \gamma^2}(t - t')] e^{(\gamma-\alpha)t'}\, dt'.$$

All the integrals are straightforward, and we obtain

$$x(t) = \frac{F_0}{\sqrt{\omega_0^2 - \gamma^2}} \frac{\sin [\sqrt{\omega_0^2 - \gamma^2}\, t - \delta]}{\sqrt{\omega_0^2 + \alpha^2 - 2\alpha\gamma}} e^{-\gamma t} + \frac{F_0}{\omega_0^2 + \alpha^2 - 2\alpha\gamma} e^{-\alpha t},$$

where $\tan \delta = \sqrt{\omega_0^2 - \gamma^2}/(\alpha - \gamma)$ with δ in the first or second quadrant. Note that the boundary conditions $x(0) = 0$, $\dot{x}(0) = 0$ are satisfied. It is also easy to check that the $x(t)$ found in this way actually does satisfy Eq. (7.25).

If we let the damping go to zero, we find that

$$x(t) = \frac{F_0}{\omega_0} \frac{\sin (\omega_0 t - \delta)}{\sqrt{\omega_0^2 + \alpha^2}} + \frac{F_0}{\omega_0^2 + \alpha^2} e^{-\alpha t}.$$

After a long time has elapsed and the force $F(t)$ has become very small, we obtain pure oscillatory motion. Thus, in the limit of no damping and of long time, we get simply

$$x(t) = \frac{F_0}{\omega_0} \frac{\sin (\omega_0 t - \delta)}{\sqrt{\omega_0^2 + \alpha^2}}.$$

From this result we can compute the final energy of the system. The system had initially zero energy, so the final energy is equal to the energy transferred to the system. Now

$$E = \tfrac{1}{2}\dot{x}^2 + \tfrac{1}{2}\omega_0^2 x^2 ,$$

for a particle of unit mass. Thus

$$E = \frac{F_0^2}{2(\omega_0^2 + \alpha^2)} .$$

According to Section 5.7,

$$E = \frac{1}{2}\left|\int_{-\infty}^{\infty} e^{-i\omega_0 t}F(t)\right|^2 = \frac{1}{2}F_0^2\left|\int_0^{\infty} e^{-i\omega_0 t}e^{-\alpha t}\,dt\right|^2 = \frac{F_0^2}{2(\omega_0^2 + \alpha^2)} ,$$

which agrees with our result.

In looking at Eq. (7.35a), one is struck by the fact that $G(t, t')$ bears a close resemblance to the solutions of the homogeneous equation

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = 0 . \tag{7.37}$$

In fact, considered as a function of t, $G(t, t')$ for $t > t'$ is a linear combination of the two linearly independent solutions of Eq. (7.37) given in Eq. (7.30). (The coefficients of this linear combination depend, of course, on t'.) For this case,

$$G(t, t') = A(t')e^{-\gamma t'}\sin[\sqrt{\omega_0^2 - \gamma^2}\,t] + B(t')e^{-\gamma t'}\cos[\sqrt{\omega_0^2 - \gamma^2}\,t] ,$$

where

$$A(t') = \frac{e^{\gamma t'}\cos[\sqrt{\omega_0^2 - \gamma^2}\,t']}{\sqrt{\omega_0^2 - \gamma^2}} , \qquad B(t') = -\frac{e^{\gamma t'}\sin[\sqrt{\omega_0^2 - \gamma^2}\,t']}{\sqrt{\omega_0^2 - \gamma^2}} .$$

The reason for this is easily seen by looking at Eq. (7.36).

Everywhere, except at $t = t'$, $G(t, t')$ must satisfy Eq. (7.37), since the δ-function vanishes except at $t = t'$. Therefore, for $t > t'$, $G(t, t')$ is some linear combination of the two solutions to Eq. (7.37) and similarly for $t < t'$.

To put the matter in a less specialized setting, let us consider the most general second-order linear differential operator

$$L \equiv f_0(t)\frac{d^2}{dt^2} + f_1(t)\frac{d}{dt} + f_2(t) .$$

Let $x_1(t)$ and $x_2(t)$ be two linearly independent solutions to

$$Lx(t) = 0 .$$

We wish to solve

$$LG(t, t') = f_0(t)\frac{d^2G(t, t')}{dt^2} + f_1(t)\frac{dG(t, t')}{dt} + f_2(t)G(t, t') = \delta(t - t') . \tag{7.38}$$

For $t > t'$, we write

$$G(t, t') = a_1x_1(t) + a_2x_2(t) , \tag{7.39a}$$

and for $t < t'$,

$$G(t, t') = b_1x_1(t) + b_2x_2(t) . \tag{7.39b}$$

Now let us try to determine a_1, a_2, b_1, and b_2. At $t = t'$, $G(t, t')$ must be continuous; if it were not, dG/dt would contain a δ-function and therefore d^2G/dt^2 would contain the *derivative* of a δ-function. However, on the right-hand side of Eq. (7.38) there is just a δ-function, so we conclude that $G(t, t')$ is continuous at $t = t'$. However, dG/dt will not be continuous at $t = t'$, and from Eq.(7.38) we can determine the value of

$$\left[\frac{dG(t, t')}{dt}\right]_{t=t'+\epsilon} - \left[\frac{dG(t, t')}{dt}\right]_{t=t'-\epsilon}$$

by integrating both sides of Eq, (7.38) from $t = t' - \epsilon$ to $t = t' + \epsilon$. On the right-hand side, the δ-function yields 1, so

$$\int_{t'-\epsilon}^{t'+\epsilon} f_0(t) \frac{d^2G(t, t')}{dt^2}\, dt + \int_{t'-\epsilon}^{t'+\epsilon} f_1(t) \frac{dG(t, t')}{dt}\, dt + \int_{t'-\epsilon}^{t'+\epsilon} f_2(t)G(t, t')\, dt = 1 .$$

Let us assume that f_0, f_1, and f_2 are continuous throughout the region of definition of the Green's function. Then since ϵ is very small, these functions will vary negligibly in the region of integration, and we can replace them by their values at $t = t'$:

$$f_0(t') \int_{t'-\epsilon}^{t'+\epsilon} \frac{d^2G(t, t')}{dt^2}\, dt + f_1(t') \int_{t'-\epsilon}^{t'+\epsilon} \frac{dG(t, t')}{dt}\, dt + f_2(t') \int_{t'-\epsilon}^{t'+\epsilon} G(t, t')\, dt = 1.$$

Since G is continuous at $t = t'$, and since the domain of integration can be made arbitrarily small, the last term on the left of this equation vanishes. Doing the remaining two integrals, we get

$$f_0(t') \left\{\left[\frac{dG(t, t')}{dt}\right]_{t=t'+\epsilon} - \left[\frac{dG(t, t')}{dt}\right]_{t=t'-\epsilon}\right\}$$
$$+ f_1(t') [G(t' + \epsilon, t') - G(t' - \epsilon, t')] = 1.$$

Since $G(t, t')$ is continuous at $t = t'$, the second term on the left-hand side vanishes and we are left with

$$\left[\frac{dG(t, t')}{dt}\right]_{t=t'+\epsilon} - \left[\frac{dG(t, t')}{dt}\right]_{t=t'-\epsilon} = \frac{1}{f_0(t')} . \tag{7.40}$$

Using this relation and the fact that $G(t, t')$ is continuous at $t = t'$, we can obtain useful information about the coefficients a_1, a_2, b_1, and b_2 in Eqs. (7.39a) and (7.39b).

From continuity at $t = t'$, we get

$$a_1x_1(t') + a_2x_2(t') = b_1x_1(t') + b_2x_2(t') ,$$

while from Eq. (7.40), we obtain

$$a_1\dot{x}_1(t') + a_2\dot{x}_2(t') - [b_1\dot{x}_1(t') + b_2\dot{x}_2(t')] = 1/f_0(t') .$$

Combining terms, we find the following set of equations for the differences $(a_1 - b_1)$ and $(a_2 - b_2)$:

$$(a_1 - b_1)x_1(t') + (a_2 - b_2)x_2(t') = 0 ,$$
$$(a_1 - b_1)\dot{x}_1(t') + (a_2 - b_2)\dot{x}_2(t') = 1/f_0(t') .$$

The solutions are

$$a_1 - b_1 = -\frac{x_2(t')}{f_0(t')\,W(t')} , \qquad a_2 - b_2 = \frac{x_1(t')}{f_0(t')\,W(t')} ,$$

where $W(t')$ is the *Wronskian* of x_1 and x_2, defined by

$$W(t') = x_1(t')\dot{x}_2(t') - \dot{x}_1(t')x_2(t')$$
$$= \begin{vmatrix} x_1(t') & x_2(t') \\ \dot{x}_1(t') & \dot{x}_2(t') \end{vmatrix} .$$

In this determinant form, the Wronskian generalizes to differential equations of any order. Putting the above result into Eqs. (7.39a, b) we see that $G(t, t')$ may be written as

$$\begin{aligned} G(t, t') &= b_1x_1(t) + b_2x_2(t) - \frac{x_1(t)x_2(t') - x_2(t)x_1(t')}{f_0(t')\,W(t')} , \qquad t > t' \\ G(t, t') &= b_1x_1(t) + b_2x_2(t) , \qquad t < t' . \end{aligned} \tag{7.41a}$$

It is left as an exercise (Problem 6) to show that $f_0(t')\,W(t')$ never vanishes. The two remaining constants, b_1 and b_2, can now be chosen to satisfy the appropriate boundary conditions. The form which the Green's function finally takes is strongly dependent on the type of boundary conditions involved. If the boundary conditions are of the single point variety typical of classical mechanics, then we shall require that $b_1 = 0 = b_2$ in Eq. (7.41a). If we do this, the Green's function is given by

$$G(t, t') = -\theta(t - t')\,\frac{x_1(t)x_2(t') - x_2(t)x_1(t')}{f_0(t')\,W(t')} , \tag{7.41b}$$

and the solution in terms of the Green's function is

$$x(t) = Ax_1(t) + Bx_2(t) + \int_{t_0}^{t} G(t, t')F(t')\,dt' . \tag{7.41c}$$

The term involving the integral vanishes at $t = t_0$ and so does its derivative with respect to t. Thus, when we apply the one-point boundary conditions, $x(t_0) = x_0$, $\dot{x}(t_0) = \dot{x}_0$, the constants A and B in Eq. (7.41c) are determined just as though the integral term were not present.

However, if the boundary conditions are of the two-point type, say

$$x(t_0) = x_0, \quad x(t_1) = x_1 ,$$

then we write

$$x(t) = Ax_1(t) + Bx_2(t) + \int_{t_0}^{t_1} G(t, t')F(t')\,dt' . \tag{7.42}$$

Then it would seem reasonable to choose b_1 and b_2 in Eq. (7.41a) so that $G(t_0, t') = 0 = G(t_1, t')$ and therefore A and B of Eq. (7.42) are determined by the boundary condition, just as though the integral term were not present.

To illustrate this case and also to demonstrate a simple calculation of a Green's function by our new method, let us consider

$$L = \frac{d^2}{dt^2} + \omega_0^2 .$$

We already know that for one-point (classical mechanical) boundary conditions, the Green's function is

$$G(t, t') = \theta(t - t') \frac{\sin[\omega_0(t - t')]}{\omega_0} ,$$

which is readily obtained by letting $\gamma = 0$ in Eq. (7.35a). Now let us use another method and apply a two-point boundary condition. We may choose

$$x_1(t) = \sin \omega_0 t , \qquad x_2(t) = \cos \omega_0 t ,$$

so that

$$W(t') = x_1(t)\dot{x}_2(t) - \dot{x}_1(t)x_2(t) = -\omega_0 .$$

Since $f_0 = 1$, we obtain from Eq. (7.41a)

$$G(t, t') = b_1 \sin \omega_0 t + b_2 \cos \omega_0 t + \frac{\sin \omega_0(t - t')}{\omega_0} , \qquad t > t' , \tag{7.43a}$$

$$G(t, t') = b_1 \sin \omega_0 t + b_2 \cos \omega_0 t , \qquad t < t' . \tag{7.43b}$$

For convenience, let us choose the endpoints to be $t_0 = 0$ and $t_1 = 1$. Then applying the condition $G(t_0, t') \equiv G(0, t') = 0$ to Eq. (7.43b), we get $b_2 = 0$. Similarly, applying the condition $G(t_1, t') \equiv G(1, t') = 0$ to Eq. (7.43a), we find that

$$b_1 \sin \omega_0 + \frac{1}{\omega_0} \sin \omega_0(1 - t') = 0 ,$$

so that

$$b_1 = \frac{1}{\omega_0} \cot \omega_0 \sin \omega_0 t' - \frac{1}{\omega_0} \cos \omega_0 t' .$$

With b_1 and b_2 evaluated, Eqs. (7.43a) and (7.43b) become

$$G(t, t') = \frac{1}{\omega_0} [\cot \omega_0 \sin \omega_0 t - \cos \omega_0 t] \sin \omega_0 t' \qquad t > t' , \tag{7.44a}$$

$$G(t, t') = \frac{1}{\omega_0} [\cot \omega_0 \sin \omega_0 t' - \cos \omega_0 t'] \sin \omega_0 t \qquad t < t' . \tag{7.44b}$$

Note how different this result is from that found for the case of one-point boundary conditions. It is interesting to note that if we let $\omega_0 \to 0$, we recover the function $G_3(t, t')$ of Eq. (7.11′) for the Green's function of d^2/dt^2 with two-point boundary conditions.

Note that when $\omega_0 = n\pi$ $(n = 1, 2, \cdots)$ in Eqs. (7.44a) and (7.44b), the function $G(t, t')$ becomes infinite and our method fails! This is because, for $\omega_0 = n\pi$,

$$L = \frac{d^2}{dt^2} + (n\pi)^2$$

has an eigenfunction with zero eigenvalue, namely, $\sin(n\pi t)$, which satisfies the boundary condition that it vanish at both $t = 0$ and $t = 1$. In this case the eigenfunction expansion of Eq. (7.18′) does not work because of a zero eigenvalue in the denominator of the expression for $G(t, t')$. And then, of course, there is *no unique* solution to the equation

$$Lx = F;$$

we can add an arbitrary multiple of $\sin(n\pi t)$ to any solution and thus get another perfectly acceptable solution.

An analysis identical to the one given here can be carried out for the general nth order linear differential operator, again with the restriction $f_0(t) \neq 0$ and with the proviso ruling out zero eigenvalues.

As was pointed out at the end of Section 7.1, the use of Green's functions is by no means restricted to solving differential equations with inhomogeneous terms which are independent of x. One of the most important applications occurs when the inhomogeneous term itself depends on the unknown function $x(t)$. For example, the differential equation

$$Lx = f(t, x)$$

can be rewritten as an integral equation of the form

$$x(t) = \xi(t) + \int G(t, t')f[t', x(t')]\, dt' ,$$

where $\xi(t)$ is a linear combination of the solutions to the homogeneous equation,

$$Lx = 0 ,$$

which satisfies the boundary conditions, and $G(t, t')$ is the Green's function belonging to L which satisfies its own appropriate boundary conditions. For example, for a simple harmonic oscillator, perturbed by some nonharmonic force of the form $\phi(t)x(t)$ or $[x(t)]^2$, we have

$$x(t) = x_0 \cos \omega_0 t + \frac{\dot{x}_0}{\omega_0} \sin \omega_0 t + \int_0^t \frac{1}{\omega_0} \sin \omega_0(t-t')f[t', x(t')]\, dt' ,$$

for the one-point boundary conditions $x(0) = x_0$, $\dot{x}(0) = \dot{x}_0$. A perturbation of the form $\phi(t)x(t)$ indicates that the "spring constant" of the harmonic oscillator is not constant in time, but instead has the form $k = k_0 + \phi(t)$.

Similarly, if the force were not perfectly harmonic, a term of the form $[x(t)]^2$ might be necessary to achieve an accurate representation of the force; this is often the case for the forces which bind solids, where small terms proportional to the square of the displacement play an important role. In the next chapter, we shall discuss methods by which such integral equations can be solved.

7.4 GREEN'S FUNCTIONS IN THREE DIMENSIONS

The method described at the end of the last chapter for obtaining Green's functions in one dimension is not generally suited for three dimensional problems. Since three dimensions corresponds to the real world, it is not surprising that the problems are harder in this case. However, there is a particularly important operator in three dimensions which yields readily to the Fourier transform method described at the beginning of the last section. The operator in question is

$$H_0 = \nabla^2 + \lambda ,$$

where ∇^2 is the familiar Laplacian operator discussed in Chapter 1. We shall employ the Fourier transform method to solve

$$H_0\psi(\mathbf{r}) = F(\mathbf{r}) , \tag{7.45}$$

assuming that $\psi(\mathbf{r})$ and $F(\mathbf{r})$ may be Fourier transformed. Let

$$\hat{F}(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int e^{-i\mathbf{k}\cdot\mathbf{r}} F(\mathbf{r}) d^3\mathbf{r} , \tag{7.46a}$$

$$\hat{\psi}(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int e^{-i\mathbf{k}\cdot\mathbf{r}} \psi(\mathbf{r}) d^3\mathbf{r} , \tag{7.46b}$$

then from Eq. (7.45),

$$\frac{1}{(2\pi)^{3/2}} \int e^{-i\mathbf{k}\cdot\mathbf{r}} \nabla^2\psi(\mathbf{r}) d^3\mathbf{r} + \lambda\hat{\psi}(\mathbf{k}) = \hat{F}(\mathbf{k}) . \tag{7.47}$$

Now we use Green's theorem, which states that

$$\int_V (F\nabla^2 G - G\nabla^2 F) d^3\mathbf{r} = \int_S (F\nabla G - G\nabla F)\cdot\mathbf{n}\, dS ,$$

where S is the surface enclosing the volume V, and $\mathbf{n}$ is the unit vector normal to S, pointing outward. Thus we conclude that

$$\frac{1}{(2\pi)^{3/2}} \int_V e^{-i\mathbf{k}\cdot\mathbf{r}} \nabla^2\psi(\mathbf{r}) d^3\mathbf{r} = \frac{1}{(2\pi)^{3/2}} \int_V \psi(\mathbf{r}) \nabla^2 e^{-i\mathbf{k}\cdot\mathbf{r}} d^3\mathbf{r}$$
$$+ \frac{1}{(2\pi)^{3/2}} \int_S [e^{-i\mathbf{k}\cdot\mathbf{r}} \nabla\psi(\mathbf{r}) - \psi(\mathbf{r}) \nabla e^{-i\mathbf{k}\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS . \tag{7.48}$$

The region of integration is the entire three-dimensional space, so in evaluating the surface integral, let us take the surface to be a sphere of radius R and take the limit as $R \to \infty$. In this case, $\mathbf{n}$ is just a unit vector in the radial direction,

so the surface term is

$$\frac{1}{(2\pi)^{3/2}}\int_S [e^{-i\mathbf{k}\cdot\mathbf{r}}\nabla\psi(\mathbf{r}) - \psi(\mathbf{r})\nabla e^{-i\mathbf{k}\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS$$
$$= \frac{1}{(2\pi)^{3/2}}\lim_{R\to\infty} R^2\left\{\int\left[e^{-i\mathbf{k}\cdot\mathbf{r}}\frac{d\psi}{dr} - \psi\frac{d}{dr}e^{-i\mathbf{k}\cdot\mathbf{r}}\right]d\Omega\right\}_{r=R}, \tag{7.48'}$$

where $d\Omega = \sin\theta\, d\theta\, d\phi$. If $\psi(\mathbf{r})$ tends to zero sufficiently rapidly as $|\mathbf{r}| \to \infty$ (roughly, if ψ vanishes faster than $1/r$), then the surface term will vanish, so Eq. (7.48) reduces to

$$\frac{1}{(2\pi)^{3/2}}\int_V e^{-i\mathbf{k}\cdot\mathbf{r}}\nabla^2\psi(\mathbf{r})d^3\mathbf{r} = -k^2\hat{\psi}(\mathbf{k}),$$

and Eq. (7.47) becomes

$$(-k^2 + \lambda)\hat{\psi}(\mathbf{k}) = \hat{F}(\mathbf{k}). \tag{7.49}$$

There are now two cases to be distinguished: $\lambda \geq 0$ and $\lambda < 0$. Let us consider the latter case first, and write $\lambda = -\kappa^2$. Then $(-k^2 - \kappa^2)$ never vanishes, so

$$\hat{\psi}(\mathbf{k}) = -\frac{\hat{F}(\mathbf{k})}{k^2 + \kappa^2}, \tag{7.50}$$

and by the Fourier integral theorem,

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) - \frac{1}{(2\pi)^{3/2}}\int\frac{\hat{F}(\mathbf{k})}{k^2 + \kappa^2}e^{i\mathbf{k}\cdot\mathbf{r}}d^3\mathbf{k},$$

where $\xi(\mathbf{r})$ represents an arbitrary linear combination of solutions of the homogeneous equation

$$H_0\phi(\mathbf{r}) = (\nabla^2 - \kappa^2)\phi(\mathbf{r}) = 0.$$

Using Eq. (7.45a), we get

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) + \int G(\mathbf{r}, \mathbf{r}')F(\mathbf{r}')d^3\mathbf{r}' \tag{7.51a}$$

with

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{(2\pi)^3}\int\frac{e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}}{k^2 + \kappa^2}d^3\mathbf{k}. \tag{7.51b}$$

Using the result of Eq. (6.51) we obtain

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi}\frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}. \tag{7.52}$$

To complete the solution, we need only determine $\xi(\mathbf{r})$ in Eq. (7.51a). In most physical problems we want $\psi(\mathbf{r})$ to remain bounded as $|\mathbf{r}|$ becomes very large. Let us see if we can solve

$$H_0\phi(\mathbf{r}) = (\nabla^2 - \kappa^2)\phi(\mathbf{r}) = 0 \tag{7.53}$$

subject to the requirement that $\phi(\mathbf{r})$ be bounded as $|\mathbf{r}| \to \infty$. Clearly, the solutions to Eq. (7.53) are just

$$\phi(\mathbf{r}) = e^{-\kappa_1 x} e^{-\kappa_2 y} e^{-\kappa_3 z}, \qquad \kappa^2 = \kappa_1^2 + \kappa_2^2 + \kappa_3^2 . \tag{7.54}$$

But all these solutions "blow up" as $|\mathbf{r}|$ becomes large, except in the directions in which $\kappa_1 x$, $\kappa_2 y$, and $\kappa_3 z$ are positive. Thus there are no acceptable solutions to the homogeneous equation, and we may write the final result as

$$\phi(\mathbf{r}) = -\frac{1}{4\pi} \int \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} F(\mathbf{r}') d^3\mathbf{r}' . \tag{7.55}$$

If $F(\mathbf{r}')$ decreases sufficiently rapidly as $|\mathbf{r}'| \to \infty$ (for example, if $F(\mathbf{r}')$ vanishes when $|\mathbf{r}'| > R$ for some R), then we see that as $|\mathbf{r}| \to \infty$,

$$\phi(\mathbf{r}) \to C \frac{e^{-\kappa r}}{r} .$$

In this case our neglect of surface terms in Eq. (7.48) was justified, since $\phi(\mathbf{r})$ drops off exponentially for large values of $|\mathbf{r}|$.

An interesting application of Eq. (7.55) occurs in electromagnetic theory, the electromagnetic potential, $\phi(\mathbf{r})$ is related to the charge distribution, $\rho(\mathbf{r})$, by Poisson's equation:

$$\nabla^2 \phi(\mathbf{r}) = -4\pi\rho(\mathbf{r}) .$$

Equation (7.55), in the limit $\kappa = 0$, tells us that

$$\phi(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' ,$$

which leads to the familiar multipole expansion for $\phi(\mathbf{r})$. For $|\mathbf{r}|$ large enough, $\phi(\mathbf{r}) \to Q/r$, where

$$Q = \int \rho(\mathbf{r}') d^3\mathbf{r}' .$$

This result is sensible, since at large enough distances any charge distribution should look like a point charge.

Another important application of Eq. (7.55) is to Schrödinger's equation,

$$\left(\nabla^2 + \frac{2mE}{\hbar^2}\right)\phi(\mathbf{r}) = \frac{2m}{\hbar^2} V(\mathbf{r})\phi(\mathbf{r}) . \tag{7.56}$$

For bound states in an attractive potential $[V(\mathbf{r}) \leq 0]$, E is negative. Hence, writing $(-2mE/\hbar^2) \equiv \kappa^2$, we get from Eq. (7.56), using Eq. (7.55),

$$\phi(\mathbf{r}) = -\frac{m}{2\pi\hbar^2} \int \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} V(\mathbf{r}')\phi(\mathbf{r}') d^3\mathbf{r}' . \tag{7.57}$$

Thus the linear, differential eigenvalue problem has been transformed into a linear, integral eigenvalue problem. Because of the complexity of the Green's function, it is often preferable to write equations like this in "momentum space," i.e., to write instead an integral equation for the Fourier transform of

$\psi(\mathbf{r})$. Such an equation is given straightaway by Eq. (7.50). For Schrödinger's equation,

$$F(\mathbf{r}) = \frac{2m}{\hbar^2} V(\mathbf{r})\psi(\mathbf{r}) ,$$

and using Eq. (7.46a) and the convolution theorem [see Eq. (5.65)], we get

$$\hat{F}(\mathbf{k}) = \frac{2m}{(2\pi)^{3/2}\hbar^2} \int \hat{V}(\mathbf{k} - \mathbf{p})\hat{\psi}(\mathbf{p})\, d^3\mathbf{p} ,$$

so Eq. (7.50) becomes

$$\hat{\psi}(\mathbf{k}) = -\frac{2m}{(2\pi)^{3/2}\hbar^2} \int \frac{\hat{V}(\mathbf{k} - \mathbf{p})}{k^2 + \kappa^2} \hat{\psi}(\mathbf{p}) d^3\mathbf{p} , \tag{7.57'}$$

which, of course, could also have been obtained by taking the Fourier transform of both sides of Eq. (7.57) and making use of the convolution theorem [Eq. (5.65)] twice.

As a specific example, suppose that $V(\mathbf{r})$ is given by the Yukawa potential

$$V(\mathbf{r}) = -g^2 \frac{e^{-\mu r}}{r} . \tag{7.58}$$

A straightforward integration shows that

$$\hat{V}(\mathbf{q}) = -\left(\frac{2}{\pi}\right)^{1/2} g^2 \frac{1}{q^2 + \mu^2} , \tag{7.58'}$$

so for this case Schrödinger's integral equation is

$$\hat{\psi}(\mathbf{k}) = \frac{mg^2}{\pi^2\hbar^2} \int \frac{\hat{\psi}(\mathbf{p})}{(k^2 + \kappa^2)[(\mathbf{k} - \mathbf{p})^2 + \mu^2]} d^3\mathbf{p} . \tag{7.59}$$

In this form it is more convenient to regard κ as fixed and g^2 to be determined as an eigenvalue; that is, it is easier to ask what potential strengths can produce a given binding energy, $\hbar^2\kappa^2/2m$, rather than what binding energies will be produced by a given potential strength, g^2. For a central potential, the wave functions have the angular dependence of the spherical harmonics. Therefore

$$\hat{\psi}(\mathbf{k}) = \phi_l(k)\, Y_{lm}(\theta_k, \phi_k) , \tag{7.60}$$

and it is a simple matter to obtain a collection of integral equations, one for each $\phi_l(k)$. For example, when $l = 0$, Y_{lm} is a constant. If we substitute Eq. (7.60) into both sides of Eq. (7.59), the angular integration is straightforward, and we get as our equation for $\phi_0(k)$,

$$\phi_0(k) = \frac{mg^2}{\pi^2\hbar^2} \int_0^\infty \frac{1}{k(k^2 + \kappa^2)} \ln\left[\frac{(k + p)^2 + \mu^2}{(k - p)^2 + \mu^2}\right] \phi_0(p)\, p\, dp . \tag{7.61}$$

One-dimensional integral equations of this kind can be solved by using techniques that will be developed in the next two chapters. Just to assure himself that this equation really does have something in common with Schrödinger's

equation, the reader who enjoys doing arduous integrations can substitute the values

$$\phi_0(p) = \frac{C}{\left(p^2 + \frac{1}{a_0^2}\right)^2}, \qquad E_0 = -\frac{\hbar^2}{2ma_0^2}, \qquad k^2 = -\frac{2mE_0}{\hbar^2} = +\frac{1}{a_0^2} \tag{7.62}$$

(where $a_0 = \hbar^2/mg^2$ is the Bohr radius of the hydrogen atom) into Eq. (7.61). Upon doing the integration and letting $\mu \to 0$ *afterward*, he will find that this $\phi_0(p)$ is in fact a solution to Eq. (7.61). [*Hint:* Integrate by parts once to get rid of the log term and then do a contour integral.] This is as it should be, since $\phi_0(p)$, as given in Eq. (7.62), is proportional to the Fourier transform of the ground state wave function of the hydrogen atom, and E_0 is the corresponding ground state energy. Note in passing that Eq. (7.61) tells us that for a Yukawa potential (and also for a Coulomb potential),

$$\phi_0(k) \sim 1/k^4$$

for k very large, as may be seen by expanding the log term in Eq. (7.61). This is true for any $l = 0$ state.

Before the reader objects that there is nothing new in all this, since everyone knows that there exist exact solutions for all the bound states of hydrogen, let us point out that the Coulomb potential is one of the very few potentials which can be dealt with in closed form. Many potentials of interest in physics do not have closed-form solutions, so other techniques, such as the ones being developed here, must be employed.

Having disposed of the case $\lambda < 0$, let us now look at the more subtle case, $\lambda \geq 0$. Now the quantity $(-k^2 + \lambda)$ in Eq. (7.49) has a zero for $k = \pm\sqrt{\lambda}$, so we cannot invert it as it stands. We shall first give a prescription for avoiding this difficulty and discuss its legitimacy later. The idea is to assume that $\sqrt{\lambda}$ is the sum of a positive real number and an imaginary number, so that

$$\lambda = (q \pm i\epsilon)^2, \qquad \epsilon > 0. \tag{7.63}$$

The use of ϵ expresses the hope that when we are finished we can let ϵ tend to zero and obtain a well-behaved result. Thus, Eq. (7.49) yields

$$\hat{\phi}_\pm(\mathbf{k}) = -\frac{\hat{F}(\mathbf{k})}{k^2 - (q \pm i\epsilon)^2}, \tag{7.64}$$

and proceeding just as before, we arrive at

$$\phi_\pm(\mathbf{r}) = \xi(\mathbf{r}) + \int G_\pm(\mathbf{r}, \mathbf{r}')F(\mathbf{r})d^3\mathbf{r}', \tag{7.65}$$

with

$$G_\pm(\mathbf{r}, \mathbf{r}') = -\frac{1}{(2\pi)^3}\int \frac{e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}}{k^2 - (q \pm i\epsilon)^2} d^3\mathbf{k}.$$

Because of the imaginary part which we have inserted, this integral has no poles on the path of integration, so we may proceed just as in the $\lambda < 0$ case.

Doing the angular integration and extending the k integral to $-\infty$ as we did in obtaining Eq. (7.52), we obtain

$$G_{\pm}(\mathbf{r}, \mathbf{r}') = -\frac{1}{8\pi^2 i|\mathbf{r} - \mathbf{r}'|}\int_{-\infty}^{\infty}\left[\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{(k - q \mp i\epsilon)(k + q \pm i\epsilon)} - \frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{(k - q \mp i\epsilon)(k + q \pm i\epsilon)}\right] k\, dk\,.$$

There are two integrals in question. We consider first

$$J_{\pm} = \int_{-\infty}^{\infty} k\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{(k - q \mp i\epsilon)(k + q \pm i\epsilon)}\, dk\,.$$

By arguments used before in similar circumstances, we can evaluate this integral by contour integration, closing the contour in the upper half-plane. Again, the contribution from the semicircle vanishes just as in the $\lambda < 0$ case, so that, by the residue theorem,

$$\begin{aligned} J_{\pm} &= 2\pi i\,\frac{(\pm q + i\epsilon)e^{i(\pm q + i\epsilon)|\mathbf{r}-\mathbf{r}'|}}{2(\pm q + i\epsilon)} \\ &= \pi i e^{\pm iq|\mathbf{r}-\mathbf{r}'|}e^{-\epsilon|\mathbf{r}-\mathbf{r}'|}\,. \end{aligned}$$

Likewise,

$$K_{\pm} = \int_{-\infty}^{\infty} k\frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{(k - q \pm i\epsilon)(k + q \pm i\epsilon)}\, dk = -\pi i e^{\pm iq|\mathbf{r}-\mathbf{r}'|}e^{-\epsilon|\mathbf{r}-\mathbf{r}'|}\,.$$

Hence

$$G_{\pm}(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi}\frac{e^{\pm iq|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}e^{-\epsilon|\mathbf{r}-\mathbf{r}'|}\,. \tag{7.66}$$

Returning to Eq. (7.65), we find that

$$\psi_{\pm}(\mathbf{r}) = \xi(\mathbf{r}) - \frac{1}{4\pi}\int\frac{e^{i(\pm q + i\epsilon)|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}F(\mathbf{r}')d^3\mathbf{r}'\,.$$

Thus the choice of positive or negative imaginary part in Eq. (7.63) has a profound effect on the solution. In one case, as $|\mathbf{r}| \to \infty$,

$$\psi_{+}(\mathbf{r}) \to \xi(\mathbf{r}) + C\frac{e^{iqr}}{r}\,,$$

whereas, in the other case,

$$\psi_{-}(\mathbf{r}) \to \xi(\mathbf{r}) + C\frac{e^{-iqr}}{r}\,.$$

In both these cases, C may depend on θ and ϕ, but not on r. Thus, in cases where the boundary condition can be put in the form of a restriction on the behavior of the solutions at large distances (such as in quantum mechanical scattering problems), the choice of $\lambda = (q + i\epsilon)^2$ or $\lambda = (q - i\epsilon)^2$ corresponds

to the application of a particular boundary condition at all points of a two-dimensional surface.

Now we must examine the possibilities for $\xi(\mathbf{r})$, the appropriate solution of the homogeneous equation

$$(\nabla^2 + q^2)\phi(\mathbf{r}) = 0 .$$

This equation has as its solution $e^{i\mathbf{q}\cdot\mathbf{r}}$, where $\mathbf{q}$ is a vector whose magnitude is q and whose direction is arbitary. Thus the complete solution reads

$$\psi_\pm(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}\cdot\mathbf{r}} - \frac{1}{4\pi}\int \frac{e^{i(\pm q + i\epsilon)|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} F(\mathbf{r}')d^3\mathbf{r}' . \tag{7.67}$$

The constant A and the direction of $\mathbf{q}$ are determined by the initial conditions.

A particularly important example is again provided by Schrödinger's equation, which we write in the form

$$\left(\nabla^2 + \frac{2mE}{\hbar^2}\right)\psi(\mathbf{r}) = \frac{2m}{\hbar^2} V(\mathbf{r})\psi(\mathbf{r}) .$$

If we look at the so-called *scattering case* ($E \geq 0$), then we have precisely the situation envisioned in deriving Eq. (7.67). We find immediately the following *integral equation* for the wave function:

$$\psi_\pm(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\int \frac{e^{\pm iq_i|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} V(\mathbf{r}')\psi_\pm(\mathbf{r}')d^3\mathbf{r}' . \tag{7.68}$$

where $q_i = \sqrt{2mE/\hbar^2}$. This equation is called the Lippmann-Schwinger equation; it is the starting point for most modern studies of quantum-mechanical scattering phenomena. We have dropped the ϵ in anticipation of the fact that the limit $\epsilon \to 0$ will pose no difficulties. We shall return to this point later.

The physical picture here is that of a plane wave traveling in the direction $\mathbf{q}_i$, with energy $\hbar^2 q_i^2/2m$, incident on a scattering center represented by $V(\mathbf{r})$. The intensity is determined by A, but is largely irrelevant to the physics of the problem. At large distances, we expect to find not only the incident wave, but also an outgoing spherical wave due to the scattering. In this case the boundary condition demands that we have an asymptotic behavior of the type

$$\psi_+(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i\cdot\mathbf{r}} + C\frac{e^{iq_i r}}{r} .$$

That this is the appropriate solution which contains *outgoing* spherical waves can be seen by remembering that Schrödinger's time-dependent equation is

$$H\Psi = i\hbar\frac{\partial\Psi}{\partial t} .$$

If H is time independent,

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-iEt/\hbar} = \psi(\mathbf{r})e^{-i\omega t} ,$$

where $\omega = E/\hbar$ and $\psi(\mathbf{r})$ satisfies Schrödinger's time-independent equation,

$$H\psi(\mathbf{r}) = E\psi(\mathbf{r}) .$$

Thus the *asymptotic* solution to the time-dependent equation is

$$\Psi_+(\mathbf{r}, t) = \frac{A}{(2\pi)^{3/2}} e^{i(\mathbf{q}_i\cdot\mathbf{r}-\omega t)} + C\frac{e^{i(q_i r-\omega t)}}{r}, \tag{7.69}$$

where $\omega = \hbar q_i^2/2m$. For the phase to remain constant as t increases, r must increase; thus Eq. (7.69) describes an *outgoing* wave.

Conversely, the asymptotic form

$$\psi_-(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i\cdot\mathbf{r}} + C\frac{e^{-iq_i r}}{r}$$

corresponds to an incoming spherical wave. This boundary condition is also relevant in quantum mechanics when discussing final states of scattering processes. Since it involves no extra difficulty to do so, we shall discuss both conditions together.

It is interesting to look a bit more closely at the asymptotic form of Eq. (7.68). If we write

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{r^2 - 2\mathbf{r}\cdot\mathbf{r}' + r'^2} = r\left[1 - \frac{2\mathbf{r}\cdot\mathbf{r}'}{r^2} + \frac{r'^2}{r^2}\right]^{1/2},$$

and assume that r is very large, then a binomial expansion yields

$$|\mathbf{r} - \mathbf{r}'| = r - \mathbf{n}\cdot\mathbf{r}' + \mathcal{O}(1/r);$$

$\mathcal{O}(1/r)$ represents terms which vanish at least as fast as $1/r$ for large r, and $\mathbf{n}$ is a unit vector in the radial direction. Thus for large r, Eq. (7.68) becomes

$$\psi_\pm(\mathbf{r}) \to \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2}\frac{e^{\pm iq_i r}}{r}\int e^{\mp iq_i\mathbf{n}\cdot\mathbf{r}'} V(\mathbf{r}')\psi_\pm(\mathbf{r}')d^3\mathbf{r}' .$$

Setting $\mathbf{q}_f = q_i\mathbf{n}$, we have

$$\psi_\pm(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}}\left[e^{i\mathbf{q}_i\cdot\mathbf{r}} + f_\pm^s(\mathbf{q}_f)\frac{e^{\pm iq_i r}}{r}\right], \tag{7.70}$$

where

$$f_\pm^s(\mathbf{q}_f) = -\frac{(2\pi)^{1/2}m}{Ah^2}\int e^{\mp i\mathbf{q}_f\cdot\mathbf{r}'} V(\mathbf{r}')\psi_\pm(\mathbf{r}')d^3\mathbf{r}'; \tag{7.71}$$

$f_\pm^s(\mathbf{q}_f)$ is called the scattering amplitude. It measures the ratio of the flux of scattered particles to that of incident particles. It is clearly independent of A, since $\psi(\mathbf{r})$ will be proportional to A. Since $\psi(\mathbf{r})$ and A have the same dimensions, we see from Eq. (7.70) that $f_\pm^s(\mathbf{q}_f)$ must have the dimensions of length. The quantity $f_\pm^s(\mathbf{q}_f)$ is of great importance in quantum mechanics, since its modulus squared is what is measured in scattering experiments. We shall soon obtain an integral equation for this important quantity. For the moment, we merely remark that since $f_\pm^s(\mathbf{q}_f)$ is, apart from constant factors, just the Fourier

transform of $V(\mathbf{r})\psi(\mathbf{r})$, we can use the convolution theorem to obtain

$$f_{\pm}^{s}(\mathbf{q}_f) = -\frac{(2\pi)^{1/2}m}{A\hbar^2}\int \hat{V}(\pm\mathbf{q}_f - \mathbf{p})\hat{\psi}(\mathbf{p})d^3\mathbf{p}\,. \tag{7.72}$$

The fact that $f_{\pm}^{s}(\mathbf{q}_f)$ is simply related to the Fourier transform of $\psi(\mathbf{r})$ suggests that it may be useful to obtain an integral equation for $\hat{\psi}(\mathbf{p})$ just as we did in Eq. (7.57′) for the bound state ($\lambda < 0$) case. The only difference is that we now have an inhomogeneous term. By looking at Eq. (7.68), we see that this must be the Fourier transform of $[A/(2\pi)^{3/2}]e^{i\mathbf{q}_i\cdot\mathbf{r}}$, which is

$$\frac{A}{(2\pi)^3}\int e^{i\mathbf{q}_i\cdot\mathbf{r}}e^{-i\mathbf{k}\cdot\mathbf{r}}d^3\mathbf{r} = A\delta(\mathbf{q}_i - \mathbf{k})\,.$$

(This representation of the δ-function was introduced in Chapter 5, Eq. 5.63.) Now proceeding exactly as we did in obtaining Eq. (7.57′), we find that

$$\hat{\psi}_{\pm}(\mathbf{k}) = A\delta(\mathbf{q}_i - \mathbf{k}) - \frac{2m}{(2\pi)^{3/2}\hbar^2}\int \frac{\hat{V}(\mathbf{k}-\mathbf{p})}{k^2 - (q_i \pm i\epsilon)^2}\hat{\psi}_{\pm}(\mathbf{p})d^3\mathbf{q}\,. \tag{7.73}$$

This is the momentum-space version of the Lippmann-Schwinger equation. Equation (7.72) suggests that we define a new function $f_{\pm}(\mathbf{q})$ by

$$f_{\pm}(\mathbf{q}) = -\frac{(2\pi)^{1/2}m}{A\hbar^2}\int \hat{V}(\mathbf{q}-\mathbf{p})\hat{\psi}_{\pm}(\mathbf{p})d^3\mathbf{p}\,. \tag{7.74}$$

Multiplying both sides of Eq. (7.73) by

$$-\frac{(2\pi)^{1/2}m}{A\hbar^2}\hat{V}(\mathbf{q}-\mathbf{k})\,,$$

and integrating over all $\mathbf{k}$-space, we obtain

$$f_{\pm}(\mathbf{q}) = -\frac{(2\pi)^{1/2}m}{\hbar^2}\hat{V}(\mathbf{q}-\mathbf{q}_i) - \frac{2m}{(2\pi)^{3/2}\hbar^2}\int \hat{V}(\mathbf{q}-\mathbf{k})\frac{f_{\pm}(\mathbf{k})}{k^2 - (q_i \pm i\epsilon)^2}d^3\mathbf{k}\,. \tag{7.75}$$

Comparing Eqs. (7.72) and (7.74), we see that the scattering amplitude is given simply by

$$f_{\pm}^{s}(\mathbf{q}_f) = f_{\pm}(\pm\mathbf{q}_f)\,,$$

where

$$\mathbf{q}_f = q_i\mathbf{n} = \sqrt{2mE/\hbar^2}\,\mathbf{n}\,.$$

The function $f_{\pm}$ has physical significance only at a particular value of $\mathbf{q}$; when we evaluate $f_{\pm}$ at this particular value, we often say that we evaluate $f_{\pm}$ "on the energy shell."

It is interesting to see what Eq. (7.75) looks like for the Yukawa potential of Eq. (7.58), whose Fourier transform is given by Eq. (7.58′). Putting this into Eq. (7.75), we get

$$f_{\pm}(\mathbf{q}) = \frac{2mg^2}{\hbar^2}\frac{1}{(\mathbf{q}-\mathbf{q}_i)^2 + \mu^2} + \frac{mg^2}{\pi^2\hbar^2}\int \frac{f_{\pm}(\mathbf{k})d^3\mathbf{k}}{[(\mathbf{q}-\mathbf{k})^2 + \mu^2][k^2 - (q_i \pm i\epsilon)^2]}\,. \tag{7.76}$$

The first term of this equation for $f_\pm$ [generally, the first term of Eq. (7.75)] is called the *Born approximation* to the scattering amplitude when evaluated at $\mathbf{q} = \pm\mathbf{q}_f$. It is not unreasonable to imagine that if the potential is weak (say, g^2 is small), then this might be a good approximation for the scattering amplitude. We shall see later in what sense this is actually the case. If we let $g^2 = e^2$, where e is the electronic charge, and set $\mu = 0$, then the Born approximation to the scattering amplitude for a Coulomb potential is

$$f^s = \frac{2me^2}{\hbar^2}\frac{1}{(\mathbf{q}_f - \mathbf{q}_i)^2}.$$

Since $|\mathbf{q}_i| = |\mathbf{q}_f|$, we may write this as

$$f^s = \frac{me^2}{\hbar^2 q_i^2}\frac{1}{1 - \cos\theta},$$

where θ is the *scattering angle*, i.e., the angle between the incident direction of the plane wave, $\mathbf{q}_i$, and the direction of the outgoing particle, $\mathbf{q}_f$. Writing $1 - \cos\theta = 2\sin^2(\theta/2)$, we have

$$f^s = \frac{me^2}{2\hbar^2 q_i^2}\frac{1}{\sin^2(\theta/2)},$$

and therefore

$$|f^s|^2 = \left(\frac{e^2}{2mv_i^2}\right)^2 \cdot \frac{1}{\sin^4(\theta/2)}, \tag{7.77}$$

since $\hbar^2 q_i^2/m = mv_i^2$, where v_i is the velocity of the incident particle. Equation (7.77) gives the relative probability of finding a particle scattered through an angle θ. It is the famous Rutherford differential cross section, first derived by Lord Rutherford from classical mechanics.

7.5 RADIAL GREEN'S FUNCTIONS

In the last section we saw how Schrödinger's momentum-space integral equation could be reduced to a one-variable integral equation by factoring out the known angular dependence of the solutions given by the spherical harmonics. This suggests that a similar reduction could be carried out at the very outset starting from Schrödinger's differential equation in position space. This can, in fact, be done, and the manner in which the Green's function is handled in this case demonstrates the relationship between the last two sections.

Writing Schrödinger's equation as

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(r)\psi(\mathbf{r}) = E\psi(\mathbf{r}),$$

where $V(r)$ depends only on the magnitude of $\mathbf{r}$, we look for solutions of the form

$$\psi(\mathbf{r}) = \phi_l(r)\, Y_{lm}(\theta, \phi).$$

Now, according to Section 5.8, the spherical harmonics are eigenfunctions of

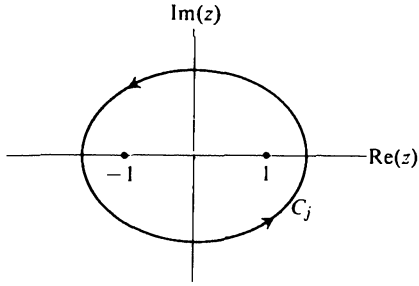

Fig. 7.5 The contour C_j for the evaluation of the spherical Bessel function.

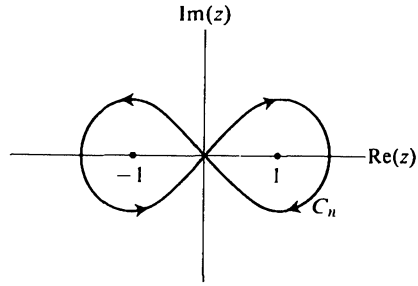

Fig. 7.6 The contour C_n for the evaluation of the spherical Neumann function.

the angular part of ∇^2 with eigenvalues $l(l+1)$. Thus $\phi_l(r)$ satisfies

$$\frac{d^2\phi_l}{dr^2} + \frac{2}{r}\frac{d\phi_l}{dr} + \frac{2mE}{\hbar^2}\phi_l(r) - \frac{l(l+1)}{r^2}\phi_l(r) = \frac{2m}{\hbar^2}V(r)\phi_l(r) . \tag{7.78}$$

Let us consider the homogeneous equation

$$\frac{d^2\phi_l}{dr^2} + \frac{2}{r}\frac{d\phi_l}{dr} + \left[\frac{2mE}{\hbar^2} - \frac{l(l+1)}{r^2}\right]\phi_l(r) = 0 . \tag{7.79}$$

If we make the change of variable

$$x = [2mE/\hbar^2]^{1/2}r , \tag{7.80}$$

the equation for ϕ_l becomes

$$\frac{d^2\phi_l}{dx^2} + \frac{2}{x}\frac{d\phi_l}{dx} + \left[1 - \frac{l(l+1)}{x^2}\right]\phi_l(x) = 0 . \tag{7.81}$$

When x tends to zero, a solution to Eq. (7.81) can tend to either x^l or x^{-l-1}. The reader should find it instructive to verify this. The regular solution which behaves like x^l near $x = 0$ is called the *spherical Bessel function* of order l, and the solution which blows up like x^{-l-1} is called the *spherical Neumann function* of order l.

Power series solutions to equations of the form of Eq. (7.81) are well known. However, we shall not proceed in this manner; instead, we shall write two linearly independent solutions in the form of complex *integral representations*. In particular, we claim that

$$j_l(x) = \frac{1}{2\pi}\frac{(-2)^l l!}{x^{l+1}}\oint_{C_j}\frac{e^{-ixz}}{(z+1)^{l+1}(z-1)^{l+1}}dz \tag{7.82}$$

is a solution to Eq. (7.81), when C_j is any contour enclosing the poles at $z = \pm 1$ in the manner shown in Fig. 7.5. Another solution is given by taking a different contour, C_n, shown in Fig. 7.6. We write

$$n_l(x) = \frac{1}{2\pi i}\frac{(-2)^l l!}{x^{l+1}}\oint_{C_n}\frac{e^{-ixz}}{(z+1)^{l+1}(z-1)^{l+1}}dz . \tag{7.83}$$

We adopt this method in order to demonstrate the power of complex variable techniques in the domain of differential equations. It is an interesting exercise to derive these integral representations starting with power series solutions corresponding to the two different types of behavior near the origin; however, at this point we content ourselves with showing that the integral representations above do indeed satisfy Eq. (7.81).

To show this, let us compute the first two derivatives of

$$f_l(x) = \frac{C_l}{x^{l+1}} \oint \frac{e^{-ixz}}{(z+1)^{l+1}(z-1)^{l+1}}\, dz\,,$$

where the contour is any closed curve containing at least one of the points $z = \pm 1$. Clearly, if it contains neither $z = +1$ nor $z = -1$, then by the Cauchy theorem, $f_l(x) \equiv 0$. We shall now show that such an $f_l(x)$ satisfies Eq. (7.81). For the first derivative we have, omitting the constant C_l,

$$f_l'(x) = -\frac{(l+1)}{x^{l+2}} \oint \frac{e^{-ixz}}{(z^2-1)^{l+1}}\, dz - \frac{i}{x^{l+1}} \oint \frac{ze^{-ixz}}{(z^2-1)^{l+1}}\, dz\,.$$

Similarly,

$$f_l''(x) = \frac{(l+1)(l+2)}{x^{l+3}} \oint \frac{e^{-ixz}\, dz}{(z^2-1)^{l+1}} + \frac{2i(l+1)}{x^{l+2}} \oint \frac{ze^{-ixz}\, dz}{(z^2-1)^{l+1}} - \frac{1}{x^{l+1}} \oint \frac{z^2 e^{-ixz}\, dz}{(z^2-1)^{l+1}}\,.$$

Now we form the quantity

$$q_l(x) = f_l''(x) + \frac{2}{x} f_l'(x) + \left[1 - \frac{l(l+1)}{x^2}\right] f_l(x)\,.$$

To show that $f_l(x)$ satisfies Eq. (7.81), we now need only show that $q_l(x) = 0$. Using the expressions for $f_l'(x)$ and $f_l''(x)$, we get

$$q_l(x) = \frac{2il}{x^{l+2}} \oint \frac{ze^{-ixz}}{(z^2-1)^{l+1}}\, dz - \frac{1}{x^{l+1}} \oint \frac{e^{-ixz}}{(z^2-1)^{l}}\, dz\,.$$

But

$$\frac{2lze^{-ixz}}{(z^2-1)^{l+1}} = -\frac{d}{dz}\left[\frac{e^{-ixz}}{(z^2-1)^{l}}\right] - ix\,\frac{e^{-ixz}}{(z^2-1)^{l}}\,,$$

so

$$q_l(x) = -\frac{i}{x^{l+2}} \oint \frac{d}{dz}\left[\frac{e^{-ixz}}{(z^2-1)^{l}}\right] dz = 0\,,$$

since the integral of an exact derivative around a closed contour vanishes.

Hence $f_l(x)$ is a solution to Eq. (7.81) for any closed contour containing a singularity. We have taken the two contours of Figs. 7.5 and 7.6 to conform to the conventional definitions. This is also the reason for our particular choice of constants multiplying the integrals in Eqs. (7.82) and (7.83). Naturally,

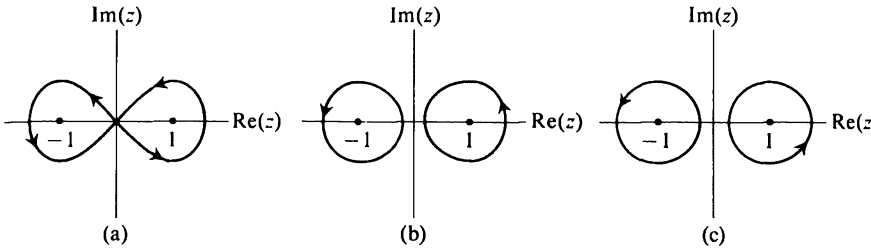

Fig. 7.7 The pinching of the contour of Figure 7-5 to obtain two contours for the evaluation of the spherical Bessel function.

the shape of the contour does not matter; Cauchy's theorem tells us that only the singularities which it encloses are relevant. For example, we have chosen for C_j a circular contour, as shown in Fig. 7.5, which contains both $z = 1$ and $z = -1$. But we can just as well pinch this contour as shown in Fig. 7.7(a), then separate the contour into two parts as shown in Fig. 7.7(b), and finally deform it into the form of *two* circular contours as shown in Fig. 7.7(c), each traced out in a counterclockwise direction, one about $z = -1$, and the other about $z = +1$. Similarly, the figure-eight contour defining $n_l(x)$ can be turned into two circular contours, the one about $z = -1$ being traced out counterclockwise, the one about $z = +1$ being traced out *clockwise*. Thus the function defined by

$$h_l^{(1)}(x) \equiv j_l(x) + in_l(x) , \tag{7.84a}$$

has the representation

$$h_l^{(1)}(x) = \frac{1}{\pi} \frac{(-2)^l l!}{x^{l+1}} \oint_{C_1} \frac{e^{-ixz}}{(z^2 - 1)^{l+1}} dz , \tag{7.84b}$$

where C_1 is a contour encircling only $z = -1$, in the counterclockwise direction. [Remember that we defined $n_l(x)$ with an extra factor of i^{-1} relative to $j_l(x)$.] Similarly,

$$h_l^{(2)}(x) \equiv j_l(x) - in_l(x) \tag{7.85a}$$

has the representation

$$h_l^{(2)}(x) = \frac{1}{\pi} \frac{(-2)^l l!}{x^{l+1}} \oint_{C_2} \frac{e^{-ixz}}{(z^2 - 1)^{l+1}} dz , \tag{7.85b}$$

where C_2 is a contour encircling only $z = 1$ in the counterclockwise sense. $h_l^{(1)}(x)$ and $h_l^{(2)}(x)$ are called the *spherical Hankel functions* of the first and second kind, respectively. Had we so desired, we could equally well have chosen $h_l^{(1)}(x)$ and $h_l^{(2)}(x)$ to be our basic solutions rather than $j_l(x)$ and $n_l(x)$.

We can now obtain the properties of $j_l(x)$ and $n_l(x)$ from the integral representations in a very simple way. These properties will in turn lead us quickly to the desired Green's function for Eq. (7.81). First of all, let us obtain ex-

plicit formulas for the two functions. Using Leibnitz's rule for differentiating the product of two functions,

$$\frac{d^n}{dx^n}[f(x)g(x)] = \sum_{m=0}^{n} \frac{n!}{m!\,(n-m)!}\left[\frac{d^{n-m}}{dx^{n-m}}f(x)\right]\frac{d^m}{dx^m}g(x)\ ,$$

along with the residue theorem for a pole of order $l+1$, [Eq. (6.17)], we find that

$$j_l(x) = \frac{i(-2)^l l!}{l!\,x^{l+1}}\sum_{m=0}^{l}\frac{l!}{m!(l-m)!}\left\{\left[\frac{d^{l-m}}{dz^{l-m}}e^{-ixz}\frac{d^m}{dz^m}\left(\frac{1}{z+1}\right)^{l+1}\right]_{z=1}\right.$$
$$\left. + \left[\frac{d^{l-m}}{dz^{l-m}}e^{-ixz}\frac{d^m}{dz^m}\left(\frac{1}{z-1}\right)^{l+1}\right]_{z=-1}\right\}.$$

The expression for $n_l(x)$ differs from the above only by a factor of i^{-1} and the presence of a minus sign in front of the term coming from the pole at $z=+1$, since this pole is encircled in a clockwise direction in the case of $n_l(x)$. The derivatives are elementary, and we find that

$$j_l(x) = \frac{i(-2)^l}{x^{l+1}}\sum_{m=0}^{l}\frac{l!}{m!(l-m)!}(-ix)^{l-m}(-1)^m\frac{(l+m)!}{l!}\left[\frac{e^{-ix}}{2^{l+m+1}} + \frac{e^{ix}}{(-2)^{l+m+1}}\right]$$
$$= \sum_{m=0}^{l}\frac{(l+m)!}{m!(l-m)!}\frac{1}{2^m}\frac{1}{x^{m+1}}i^{l-m+1}\left[\frac{(-1)^{-(l-m+1)}e^{ix}+e^{-ix}}{2}\right],$$

where we have several times made use of the fact that $(-1)^m = (-1)^{-m}$. Writing $(-1) = e^{i\pi}$, $i = e^{i\pi/2}$, we get finally

$$j_l(x) = \sum_{m=0}^{l}\frac{(l+m)!}{m!(l-m)!}\frac{1}{2^m}\frac{\cos\left(x - \frac{l-m+1}{2}\pi\right)}{x^{m+1}}\ . \tag{7.86a}$$

Changing the sign of e^{-ix} in the previous expression and dividing by i, we get

$$n_l(x) = \sum_{m=0}^{l}\frac{(l+m)!}{m!(l-m)!}\frac{1}{2^m}\frac{\sin\left(x - \frac{l-m+1}{2}\pi\right)}{x^{m+1}}\ . \tag{7.86b}$$

According to our definitions of the spherical Hankel functions, we have immediately

$$h_l^{(1)}(x) = \sum_{m=0}^{l}\frac{(l+m)!}{m!(l-m)!}\frac{1}{2^m}\frac{\exp\left(i\left[x - \frac{l-m+1}{2}\pi\right]\right)}{x^{m+1}}\ , \tag{7.86c}$$

$$h_l^{(2)}(x) = \sum_{m=0}^{l}\frac{(l+m)!}{m!(l-m)!}\frac{1}{2^m}\frac{\exp\left(-i\left[x - \frac{l-m+1}{2}\pi\right]\right)}{x^{m+1}}\ . \tag{7.86d}$$

Thus, after a few simple manipulations, we have explicit expressions for all the spherical Bessel functions and spherical Hankel functions. Let us now examine their asymptotic behavior. First of all, it is clear that as x tends toward infinity, the dominant term in each sum will be $m = 0$, so

$$j_l(x) \xrightarrow[x\to\infty]{} \frac{\cos\left(x - \frac{l+1}{2}\pi\right)}{x}, \tag{7.87a}$$

$$n_l(x) \xrightarrow[x\to\infty]{} \frac{\sin\left(x - \frac{l+1}{2}\pi\right)}{x}, \tag{7.87b}$$

$$h_l^{(1)}(x) \xrightarrow[x\to\infty]{} \frac{\exp\left(i\left[x - \frac{l+1}{2}\pi\right]\right)}{x}, \tag{7.87c}$$

$$h_l^{(2)}(x) \xrightarrow[x\to\infty]{} \frac{\exp\left(-i\left[x - \frac{l+1}{2}\pi\right]\right)}{x}. \tag{7.87d}$$

As $x \to 0$, the term $m = l$ in the sum for $n_l(x)$ dominates, and we find that

$$n_l(x) \xrightarrow[x\to 0]{} \frac{(2l)!}{2^l l!} \frac{\sin(x - \pi/2)}{x^{l+1}}.$$

Thus

$$n_l(x) \xrightarrow[x\to 0]{} -\frac{(2l)!}{2^l l!} \frac{1}{x^{l+1}}, \tag{7.88a}$$

since $\sin(x - \pi/2) = -\cos x$, which tends toward -1 as x tends toward zero.

Now, at first glance it would appear reasonable to do the same thing for $j_l(x)$. If we proceed in this manner, we are rapidly led to

$$j_l(x) \xrightarrow[x\to 0]{} \frac{(2l)!}{2^l l!} \frac{\cos(x - \pi/2)}{x^{l+1}}.$$

But $\cos(x - \pi/2) = \sin x$, and since $\sin x$ varies like x as x tends to zero, we find that $j_l(x)$ appears to vary like x^{-l}. Before we conclude that this is the limiting behavior, we must check the term $m = l - 1$ in the sum for $j_l(x)$. For x very small this term varies like

$$\frac{(2l-1)!}{(l-1)!\,2^{l-1}} \frac{\cos(x - \pi)}{x^l} = -\frac{(2l-1)!}{(l-1)!\,2^{l-1}} \frac{1}{x^l}.$$

However,

$$\frac{(2l-1)!}{(l-1)!\,2^{l-1}} = \frac{(2l)!}{l!\,2^l},$$

so this term ($m = l - 1$) exactly cancels the leading contribution from the term $m = l$. That this should be the case is really not surprising, since we have seen earlier that the two solutions to Eq. (7.91) vary like either x^{-l-1} or x^l. We have seen that $n_l(x)$ varies like x^{-l-1}, so we suspect that if all is well, $j_l(x)$ should vary like x^l for small x. Hence the expression Eq. (7.86a) for $j_l(x)$ is rather misleading as far as small x behavior is concerned. This is not to say that Eq. (7.86a) is incorrect, but merely that when x becomes small there must be a great deal of cancellation (the cancellation increases rapidly with l) among the various terms contributing to $j_l(x)$, which ultimately produces the desired x^l behavior for small x.

To see that $j_l(x)$ actually does vary like x^l for small x let us return to the integral representation, Eq. (7.82), and make the change of variable $\xi = xz$. Then

$$j_l(x) = \frac{1}{2\pi} \frac{(-2)^l l!}{x^{l+2}} \oint_{C_j} \frac{e^{-i\xi}}{(\xi^2/x^2 - 1)^{l+1}} d\xi ,$$

where C_j is any counterclockwise contour enclosing the singularities at $\xi = \pm x$. We rewrite this as

$$j_l(x) = \frac{1}{2\pi} (-2)^l l! x^l \oint_{C_j} \frac{e^{-i\xi}}{\xi^{2l+2}} \left[1 - \frac{x^2}{\xi^2}\right]^{-l-1} d\xi .$$

But in any closed region outside the circle of radius $|\xi| = x$, the quantity in square brackets has a uniformly convergent power series:

$$\left[1 - \frac{x^2}{\xi^2}\right]^{-l-1} = 1 + (l+1)\frac{x^2}{\xi^2} + \frac{1}{2}(l+1)(l+2)\frac{x^4}{\xi^4} + \cdots$$

$$= \sum_{m=0}^{\infty} \frac{(l+m)!}{m!\,l!} \frac{x^{2m}}{\xi^{2m}} .$$

Since we are free to do so, let us choose C_j to be a circle lying outside the disc $\xi = x$. Because of uniform convergence, we may interchange summation and integration to obtain

$$j_l(x) = \frac{1}{2\pi} (-2)^l l! x^l \left[\oint_{C_j} \frac{e^{-i\xi}}{\xi^{2l+2}} d\xi + (l+1)x^2 \oint_{C_j} \frac{e^{-i\xi}}{\xi^{2l+4}} d\xi + \cdots\right]$$

$$= i(-2)^l l! x^l \left[\frac{(-i)^{2l+1}}{(2l+1)!} + (l+1)\frac{(-i)^{2l+3}}{(2l+3)!} x^2 + \cdots\right]$$

by a simple application of the residue calculus. Thus

$$j_l(x) = \frac{2^l l!}{(2l+1)!}\left[x^l - \frac{x^{l+2}}{2(2l+3)} + \cdots\right]. \tag{7.88b}$$

The leading term is proportional to x^l, as we had anticipated. We leave it to the interested reader to show that if one keeps all terms in the expansion of

$[1-x^2/\xi^2]^{-l-1}$, then

$$j_l(x) = 2^l x^l \sum_{m=0}^{\infty} (-1)^m \frac{(l+m)!\,x^{2m}}{m!(2l+2m+1)!}\,.$$

$$= \frac{\sqrt{\pi}}{2} \sum_{m=0}^{\infty} (-1)^m \frac{(x/2)^{2m+l}}{m!\,\Gamma(l+m+3/2)}\,, \tag{7.88b'}$$

where we have used Eq. (6.54) as the definition of $\Gamma(x)$. It follows readily from Eq. (6.54) [see, for example, the derivation of Eq. (6.56)] that

$$\Gamma(n+3/2) = \sqrt{\pi}\,\frac{(2n+1)!}{2^{2n+1}\,n!}\,,$$

which result has been used in obtaining Eq. (7.88b').

If the reader will glance back at Eq. (6.69), he will note a rather strong similarity between the above expression for $j_l(x)$ and the expression for $J_n(x)$ given there. For integral values of m and n, we know that $(m+n)! = \Gamma(m+n+1)$, and once this substitution is made, it is apparent that Eq. (6.69) generalizes to non-integral values of n. Then using Eq. (7.88b'), it takes only a few steps to show that

$$j_l(x) = \sqrt{\frac{\pi}{2x}}\,J_{l+1/2}(x)\,.$$

This is why the functions $j_l(x)$ are called spherical *Bessel* functions; they are closely related to the Bessel functions of half-integral order.

With exact expressions and all the asymptotic information for the various solutions to Eq. (7.81), we are finally ready to determine the Green's function for Eq. (7.79). Let us take the two linearly independent solutions of Eq. (7.81) to be $j_l(x)$ and $n_l(x)$. The corresponding solutions to Eq. (7.79) are $j_l(kr)$ and $n_l(kr)$, where we assume first that $E \geq 0$ and define, according to Eq. (7.80),

$$k \equiv \sqrt{2mE/\hbar^2}\,.$$

Thus, according to Eq. (7.41), the Green's function for Eq. (7.79) is

$$g_l(r, r') = b_1 j_l(kr) + b_2 n_l(kr) - \frac{1}{W(r')}\,[j_l(kr)n_l(kr') - j_l(kr')n_l(kr)\,] \tag{7.89a}$$

if $r > r'$, and

$$g(r, r') = b_1 j_l(kr) + b_2 n_l(kr) \tag{7.89b}$$

if $r < r'$. $W(r')$ is the Wronskian, given by

$$W(r') = j_l(kr')\,\frac{d}{dr'}\,n_l(kr') - n_l(kr')\,\frac{d}{dr'}\,j_l(kr')\,.$$

In the terminology of Section 7.3 [Eq. (7.38)], $f_0(r) = 1$, $f_1(r) = 2/r$, $f_2(r) = k^2 - l(l+1)r^2$. We saw in Section 7.3 that W satisfies the equation

$$\frac{dW}{dr'} = -\frac{f_1(r')}{f_0(r')}\,W\,.$$

Substituting for f_0 and f_1, we find that

$$\frac{dW}{dr'} = -\frac{2}{r'}W ,$$

from which we conclude that $W(r') = C(1/r'^2)$.

Now we must determine C, which depends on the particular normalization of the functions. Since we know the asymptotic form of j_l and n_l at large distances, we can use this knowledge to determine $W(r')$ for large r', and hence C. Using Eqs. (7.87a) and (7.87b), we have

$$W(r') \xrightarrow[r'\to\infty]{} \frac{1}{kr'^2} ,$$

from which we conclude that $C = 1/k$. Of course, the identical result could be obtained by using the expressions for small r'. Thus, quite generally,

$$W(r') = 1/kr'^2 ,$$

and Eqs. (7.89a) and (7.89b) become

$$g_l(r, r') = b_1 j_l(kr) + b_2 n_l(kr) - kr'^2[j_l(kr)n_l(kr') - j_l(kr')n_l(kr)] \tag{7.90a}$$

for $r > r'$, and

$$g_l(r, r') = b_1 j_l(kr) + b_2 n_l(kr) \tag{7.90b}$$

for $r < r'$. Now we do not want the solutions to Schrödinger's equation to be singular at the origin, so for $r < r'$, $g_l(r, r')$ must have no singularity at $r = 0$. However, $n_l(kr)$ has a singularity of the type $(1/r)^{l+1}$ at $r = 0$, so we must exclude the term in n_l from the solution. Hence we take $b_2 = 0$. For large r, a possible boundary condition is to require that the solution have the form e^{ikr}/r, as we did in the last section. But this is the asymptotic form of the function $h_l^{(1)}$, so let us choose b_1 so as to ensure this behavior for $r > r'$. If we take

$$b_1 = -ikr'^2 h_l^{(1)}(kr') ,$$

then for $r > r'$ we have, using Eq. (7.90a) with $b_2 = 0$ and Eq. (7.84a),

$$g_l(r, r') = -ikr'^2 h_l^{(1)}(kr) j_l(kr') .$$

For $r < r'$, this value for b_1 leads to the result

$$g_l(r, r') = -ikr'^2 j_l(kr) h_l^{(1)}(kr') .$$

This completes the specification of g_l. Note that $g_l(r, r')$ is continuous at $r = r'$, although, of course, its derivatives are not. One often introduces the notation $r_> \equiv$ larger of (r, r') and $r_< \equiv$ smaller of (r, r'). Using this convention, we have the following simple expression for g_l:

$$g_l(r, r') = -ikr'^2 j_l(kr_<) h_l^{(1)}(kr_>) . \tag{7.91}$$

Now, if we write Schrödinger's equation as

$$\frac{d^2\phi_l}{dr^2} + \frac{2}{r}\frac{d\phi_l}{dr} + \left[k^2 - \frac{l(l+1)}{r^2}\right]\phi_l(r) = \frac{2m}{\hbar^2}V(r)\phi_l(r) ,$$

then by our previous work we can immediately transform this into the integral equation

$$\phi_l(r) = Aj_l(kr) + \frac{2m}{\hbar^2}\int_0^\infty g_l(r, r')\, V(r')\phi_l(r')\, dr' . \tag{7.92}$$

Or, more explicitly,

$$\phi_l(r) = Aj_l(kr) - \frac{2mki}{\hbar^2}\int_0^\infty j_l(kr_<)h_l^{(1)}(kr_>)\, V(r')\phi_l(r')r'^2\, dr' . \tag{7.93}$$

We have added an arbitrary multiple of $j_l(kr)$, that solution of the homogeneous equation which is regular at the origin.

In our study of Schrödinger's equation, we have considered only the case $E \geq 0$, but it is easy to remedy this by defining

$$\kappa^2 \equiv -2mE/\hbar^2$$

and writing Schrödinger's equation as

$$\frac{d^2\phi_l}{dr^2} + \frac{2}{r}\frac{d\phi_l}{dr} + \left[-\kappa^2 - \frac{l(l+1)}{r^2}\right]\phi_l(r) = \frac{2m}{\hbar^2}V(r)\phi_l(r) .$$

If we now choose the square root of $-\kappa^2$ with *positive* imaginary part, then the solutions to the zero-order equation are $j_l(i\kappa r)$ and $n_l(i\kappa r)$. In this case, $h_l^{(1)}(i\kappa r)$ has just the type of behavior at large distances that we want in a bound state problem, namely, exponential decrease. All other solutions grow exponentially at large distances, so our integral equation becomes

$$\phi_l(r) = \frac{2m\kappa}{\hbar^2}\int_0^\infty j_l(i\kappa r_<)h_l^{(1)}(i\kappa r_>)\, V(r')\phi_l(r')r'^2\, dr' . \tag{7.94}$$

There is no inhomogeneous term present because there is no solution to the equation

$$\frac{d^2\phi_l}{dr^2} + \frac{2}{r}\frac{d\phi_l}{dr} + \left[-\kappa^2 - \frac{l(l+1)}{r^2}\right]\phi_l(r) = 0,$$

which satisfies the boundary conditions of regularity at $r = 0$ and exponential decrease at large distances.

It is interesting to compare these one-dimensional integral equations, for example Eq. (7.94), with the three-dimensional results of the last section. In the bound state case, we found in Section 7.4, Eq. (7.57), that

$$\psi(\mathbf{r}) = -\frac{m}{2\pi\hbar^2}\int \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}V(\mathbf{r}')\psi(\mathbf{r}')\, d^3r' . \tag{7.57}$$

Since the Green's function depends only on $|\mathbf{r} - \mathbf{r}'|$, we can consider it as a function of r, r', and x, where x is the cosine of the angle between $\mathbf{r}$ and $\mathbf{r}'$. It is natural to expand it in a series of Legendre polynomials in x. To this end, we define

$$g_l(r, r') \equiv \tfrac{1}{2}\int_{-1}^{1} \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}P_l(x)\, dx . \tag{7.95}$$

Then

$$\frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} = \sum_{l=0}^{\infty} (2l+1) g_l(r, r') P_l(x) , \tag{7.96}$$

where we have used the fact [Eq. (5.37), Theorem 5.3] that

$$\int_{-1}^{1} P_l(x) P_{l'}(x)\, dx = \frac{2}{2l+1}\, \delta_{ll'} .$$

We now insert Eq. (7.96) into Eq. (7.57) to get

$$\psi(\mathbf{r}) = -\frac{m}{2\pi\hbar^2} \int \sum_{l=0}^{\infty} (2l+1) g_l(r, r') P_l(x) V(r') \psi(\mathbf{r}')\, d^3r' .$$

Next we use the addition theorem for spherical harmonics, Eq. (5.83), which tells us that if x is the cosine of the angle between $\mathbf{r}$ and $\mathbf{r}'$ and if the orientation of $\mathbf{r}$ is specified by (θ, ϕ) and that of $\mathbf{r}'$ is specified by (θ', ϕ'), then

$$P_l(x) = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}(\theta, \phi)\, Y_{lm}^*(\theta', \phi') .$$

Thus

$$\psi(\mathbf{r}) = -\frac{2m}{\hbar^2} \int \sum_{l=0}^{\infty} \sum_{m=-l}^{l} g_l(r, r')\, Y_{lm}(\theta, \phi)\, Y_{lm}^*(\theta', \phi')\, V(r') \psi(\mathbf{r}')\, d^3r' . \tag{7.97}$$

We now look for solutions of the form

$$\psi(\mathbf{r}) = \phi_l(r)\, Y_{lm}(\theta, \phi) .$$

Inserting this into Eq. (7.97) and using the completeness and orthonormality of the spherical harmonics, we get

$$\phi_l(r) = -\frac{2m}{\hbar^2} \int_0^{\infty} g_l(r, r')\, V(r') \phi_l(r') r'^2\, dr' ,$$

since $V(r')$ depends only on the magnitude of $\mathbf{r}'$.

It is possible to show that $g_l(r, r')$, as defined by Eq. (7.95), is equal to $-\kappa j_l(i\kappa r_<) h_l^{(1)}(i\kappa r_>)$, so that the two approaches agree completely. This result follows from the relation

$$\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} = ik \sum_{l=0}^{\infty} (2l+1) j_l(kr_<) h_l^{(1)}(kr_>) P_l(x) \tag{7.98}$$

and Eq. (7.95). This is a very useful relationship in mathematical physics, but it would not be instructive to carry out an independent demonstration of it here. [One could, of course, regard the work of this section as a derivation of Eq. (7.98).] We content ourselves with showing that for $l = 0$, $g_l(r, r')$ as defined by Eq. (7.95) is indeed equal to $-\kappa j_l(i\kappa r_<) h_l^{(1)}(i\kappa r_>)$. To do this, we must evaluate

$$g_0(r, r') = \tfrac{1}{2} \int_{-1}^{1} \frac{e^{-\kappa|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} P_0(x)\, dx .$$

We write

$$u = |\mathbf{r} - \mathbf{r}'| = (r^2 + r'^2 - 2rr' \cos\theta)^{1/2} = (r^2 + r'^2 - 2rr'x)^{1/2} .$$

Thus

$$du = -\frac{rr'\,dx}{u}, \qquad dx = -\frac{u\,du}{rr'} .$$

Making use of this change of variable, we find that

$$g_0(r, r') = \frac{1}{2rr'} \int_{|r-r'|}^{|r+r'|} e^{-\kappa u}\,du = -\frac{1}{2rr'\kappa} [e^{-\kappa|r+r'|} - e^{-\kappa|r-r'|}] .$$

In terms of the variables $r_>$ and $r_<$, this is

$$\begin{aligned} g_0(r, r') &= -\frac{1}{2\kappa} \frac{e^{-\kappa(r_> + r_<)} - e^{-\kappa(r_> - r_<)}}{r_> r_<} \\ &= \kappa \frac{e^{-\kappa r_>}}{\kappa r_>} \left[\frac{e^{\kappa r_<} - e^{-\kappa r_<}}{\kappa r_<}\right] \\ &= \kappa \left[\frac{e^{-\kappa r_>}}{\kappa r_>}\right] \left[\frac{\sinh(\kappa r_<)}{\kappa r_<}\right] . \end{aligned}$$

But by Eqs. (7.86a) and (7.86c),

$$j_0(i\kappa r_<) = \frac{\sin(i\kappa r_<)}{i\kappa r_<} \equiv \frac{\sinh(\kappa r_<)}{\kappa r_<}, \qquad h_0^{(1)}(i\kappa r_>) = -\frac{e^{-\kappa r_>}}{\kappa r_>},$$

so

$$g_0(r, r') = -\kappa j_0(i\kappa r_<) h_0^{(1)}(i\kappa r_>) .$$

Thus there is a nice relationship between the techniques of Sections 7.3 and 7.4 whenever the potential in Schrödinger's equation is spherically symmetric. Depending on the type of problem, the three-dimensional approach or the one-dimensional approach may be more useful. Both of them have been used extensively in quantum theoretical studies of scattering processes.

At this juncture, it is perhaps worthwhile to emphasize again the importance of boundary conditions in determining the form of a Green's function. In the work of this section we have reduced ∇^2 to a one-dimensional operator by factoring out its angular dependence and have thereby seen how to apply in a straightforward way the boundary conditions appropriate to the scattering problem. The "$i\epsilon$-prescription" used in Section 7.4 is a very convenient shorthand device for achieving the same end. Its great practical utility more than justifies its use, but it is good to remember that it is in one sense a "trick." In considering a general three-dimensional operator it is often impossible to find such a prescription, and one is in trouble unless the operator in question can be brought to an effective one-dimensional form, where the methods of Section 7.3 can be used.

In this context, however, it should also be pointed out that whenever one can obtain an object which satisfies

$$\mathcal{O}G(x, x') = \delta(x - x') ,$$

then if one also has the solutions to $\mathcal{O}\psi = 0$, the general solution to

$$\mathcal{O}\psi(x) = F(x)$$

can be written as

$$\psi(x) = \xi(x) + \int G(x, x')F(x')\,dx' .$$

However, there is an important difference between this formal solution and the very similar equations written in previous sections. There we imagined that the coefficients of the solutions to $\mathcal{O}\psi = 0$ appearing in $\xi(x)$ were *independent of* $F(x)$, the inhomogeneous term. In fact, strictly speaking, this is what we meant by a Green's function; it was a function which enabled us to solve a given operator equation in such a way that the coefficients occurring in $\xi(x)$ could be determined without having to think about the inhomogeneous term of the operator equation. However, if all we know about $G(x, x')$ is that it satisfies $\mathcal{O}G(x, x') = \delta(x - x')$, then it will, in general, be the case that the coefficients in $\xi(x)$ will depend on F. This is not surprising, since to any solution of $\mathcal{O}G(x, x') = \delta(x - x')$ we can add an arbitrary linear combination of solutions to $\mathcal{O}\psi(x) = 0$, so the equation $\mathcal{O}G(x, x') = \delta(x - x')$ does not uniquely specify the function $G(x, x')$; that is, one must specify definite boundary conditions for $G(x, x')$.

For example, suppose that for some reason the only function satisfying

$$(\nabla^2 + k^2)G(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}')$$

which was known to us was the function

$$G_-(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi}\frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} . \tag{7.99a}$$

Then we could certainly write Schrödinger's equation,

$$(\nabla^2 + k^2)\psi(x) = \frac{2m}{\hbar^2}V(x)\psi(x) ,$$

as

$$\psi(\mathbf{r}) = \xi(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int G_-(\mathbf{r}, \mathbf{r}')\,V(\mathbf{r}')\psi(\mathbf{r}')d^3r' . \tag{7.99b}$$

But if we want the function $\psi(\mathbf{r})$ to behave at large values of $|\mathbf{r}|$ like

$$\psi(\mathbf{r}) \sim e^{i\mathbf{k}\cdot\mathbf{r}} + f(\theta, \phi)\frac{e^{+ikr}}{r} ,$$

that is, if we want the potential to produce only *outgoing* spherical waves, then the function $\xi(r)$ needed to satisfy the boundary conditions would not be simply $e^{i\mathbf{k}\cdot\mathbf{r}}$ (then we would have just *incoming* spherical waves), but rather $e^{i\mathbf{k}\cdot\mathbf{r}}$ *plus* a complicated linear combination of solutions to the homogeneous equation,

$$(\nabla^2 + k^2)\phi(\mathbf{r}) = 0 ,$$

whose coefficients would depend on $V(\mathbf{r})$. For this purpose, it would be convenient to take the solutions to be of the form

$$\phi_{lm}(\mathbf{r}) = j_l(kr)\, Y_{lm}(\theta, \phi) ,$$

where we reject solutions depending on $n_l(kr)$ because they are singular at the origin. (Note that $e^{i\mathbf{k}\cdot\mathbf{r}}$ is a particular linear combination of these $\phi_{lm}(\mathbf{r})$. What are the coefficients of this linear combination? See Problem 7.14.) The reader should show (Problem 7.15) that if we take

$$\xi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{4imk}{\hbar^2} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} j_l(kr)\, Y_{lm}(\theta, \phi)$$
$$\times \int Y_{lm}^*(\theta', \phi') j_l(kr')\, V(\mathbf{r}')\psi(\mathbf{r}')\, d\mathbf{r}' ,$$

then Eq. (7.99b) reduces to the familiar result

$$\psi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2} \int \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} V(\mathbf{r}')\psi(\mathbf{r}')\, d\mathbf{r}' .$$

This is just the result of Eq. (7.68) with the proper sign in the exponential to guarantee *outgoing* spherical waves. (Naturally, we could also view the above situation in terms of adding an appropriate linear combination of terms to the "incoming spherical wave Green's function" of Eq. (7.99a) in order to turn it into the desired "outgoing spherical wave Green's function.")

Thus we see that even if one starts with a Green's function satisfying very inconvenient boundary conditions, it is possible to utilize the freedom in the choice of $\xi(r)$ (or equivalently the freedom to add solutions of the homogeneous equation to $G(x, x')$) to overcome this difficulty. It should be clear, however, that in three dimensions this procedure would, in general, present very serious practical difficulties. Fortunately, many of the most important operators of physics are sufficiently simple in form so that it is a relatively straightforward matter to obtain for them Green's functions which satisfy physically useful boundary conditions. However, it is also often the case that even for simple operators the boundary conditions in three (or even two) dimensions are so complicated that the determination of the Green's function becomes virtually impossible. The following section is devoted to such a problem.

7.6 AN APPLICATION TO THE THEORY OF DIFFRACTION

In many problems of practical interest in physics, the boundary conditions are quite complicated from a mathematical point of view, even though the physical situation may be comparatively simple; this means that the appropriate operator equation is made difficult by the form of the boundary condition, rather than by the complexity of the operator itself. This is typically the case in electromagnetic theory where the presence of conducting surfaces—often not even "perfectly conducting" surfaces—makes the solution of many realistic problems extremely difficult, even after the most convenient geometrical configura-

tions have been chosen. However, because of the relatively simple form of the *operator*, it is often possible to solve the Green's function equation,

$$LG(\mathbf{r}, \boldsymbol{\rho}) = \delta^3(\mathbf{r} - \boldsymbol{\rho}) , \tag{7.100}$$

subject to some simple boundary condition. This would be the case in electromagnetic theory, for example, where $L = \nabla^2 + k^2$; if we then ask for the outgoing spherical wave solution, we get the Green's function of Eq. (7.66).

It is thus natural to ask the question: Can a solution to Eq. (7.100) help us obtain a solution to the equation,

$$L\psi = \phi , \tag{7.101}$$

where ψ must satisfy much more complicated boundary conditions than those imposed on $G(\mathbf{r}, \boldsymbol{\rho})$ in Eq. (7.100)?

To answer this question, let us first make the assumption that the operator L has the form

$$\nabla \cdot [f(\mathbf{r})\nabla] + g(\mathbf{r}) , \tag{7.102}$$

which is sufficiently general for many physical applications. Now, if ψ is any solution of Eq. (7.101), and if $G(\mathbf{r}, \boldsymbol{\rho})$ satisfies Eq. (7.100), then

$$\psi(\boldsymbol{\rho}) = \int_V \phi(\mathbf{r})G(\mathbf{r}, \boldsymbol{\rho})d^3\mathbf{r} + \int_V [\psi(\mathbf{r})LG(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})L\psi(\mathbf{r})]\, d^3\mathbf{r} , \tag{7.103}$$

where V is the volume throughout which we want the solution. Using the specific form of L [Eq. (7.102)], we can rewrite Eq. (7.103) as

$$\psi(\boldsymbol{\rho}) = \int_V \phi(\mathbf{r})G(\mathbf{r}, \boldsymbol{\rho})d^3\mathbf{r} + \int_V [\psi(\mathbf{r})\nabla \cdot f(\mathbf{r})\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla \cdot f(\mathbf{r})\nabla\psi(\mathbf{r})]\, d^3\mathbf{r} .$$

From the properties of the del operator, we find that

$$\psi(\boldsymbol{\rho}) = \int_V \phi(\mathbf{r})G(\mathbf{r}, \boldsymbol{\rho})d^3\mathbf{r} + \int_V \nabla \cdot [\psi(\mathbf{r})f(\mathbf{r})\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})f(\mathbf{r})\nabla\psi(\mathbf{r})]\, d^3\mathbf{r} .$$

By Gauss's theorem, this last expression can be written as an integral over the surface S which encloses V. Denoting the outward normal to S by $\mathbf{n}$, we have

$$\psi(\boldsymbol{\rho}) = \int_V \phi(\mathbf{r})G(\mathbf{r}, \boldsymbol{\rho})d^3\mathbf{r} + \int_S f(\mathbf{r})[\psi(\mathbf{r})\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla\psi(\mathbf{r})] \cdot \mathbf{n}\, dS . \tag{7.104}$$

Since $\phi(\mathbf{r})$, $f(\mathbf{r})$ and $G(\mathbf{r}, \boldsymbol{\rho})$ are presumed known, all that is required for the evaluation of the integrals is the value of ψ and its normal derivative *on the surface S*.

It must be emphasized, however, that the situation is really more complicated than Eq. (7.104) suggests because $\psi(\mathbf{r})$ and its normal derivative are not *both* arbitrary on S. As a boundary condition we might have $\psi(\mathbf{r})$ specified on S *or* $\nabla\psi(\mathbf{r}) \cdot \mathbf{n}$ specified on S, or even a combination of $\psi(\mathbf{r})$ and $\nabla\psi(\mathbf{r}) \cdot \mathbf{n}$ specified, *but not both separately*. This is a familiar situation from one-dimensional problems, where for second-order equations we specify the value of the

function at the endpoints, not both the value of the function and its first derivative at both endpoints.

In many circumstances, this difficulty can be overcome by making use of the freedom of choice of $G(\mathbf{r}, \boldsymbol{\rho})$ which arises from the fact that to any solution of Eq. (7.100) we can add an arbitrary multiple of any solution of $L\xi(\mathbf{r}) = 0$. If, for example, we can find a $G(\mathbf{r}, \boldsymbol{\rho})$ which satisfies Eq. (7.100) and also vanishes when $\mathbf{r}$ is on the surface, S, then Eq. (7.104) takes the form

$$\psi(\boldsymbol{\rho}) = \int_V \phi(\mathbf{r})G(\mathbf{r}, \boldsymbol{\rho})d^3\mathbf{r} + \int_S f(\mathbf{r})\psi(\mathbf{r})\nabla G(\mathbf{r}, \boldsymbol{\rho}) \cdot \mathbf{n}\, dS.$$

Thus, if the boundary condition involves a specification of $\psi(\mathbf{r})$ on S the problem is formally solved. This is a very common problem in electrostatics [$f(\mathbf{r}) \equiv 1$, $g(\mathbf{r}) \equiv 0$ in Eq. (7.102)], where numerous special techniques are available for the solution of Eq. (7.100) subject to the requirement that $G(\mathbf{r}, \boldsymbol{\rho})$ vanish on S. For a discussion of some of these methods, the reader is referred to Jackson's excellent book, *Classical Electrodynamics*. Notice that Eq. (7.100) is, apart from factors of 4π, just Poisson's equation for the potential due to a point charge at $\boldsymbol{\rho}$. Thus the above technique applied to electrostatics boils down to just the replacement of Poisson's equation for an arbitrary charge distribution and some complicated restriction on the potential at certain boundaries by Poisson's equation for a point charge and a simple boundary condition. (Note that if the shape of S is complicated even the boundary condition that $G(\mathbf{r}, \boldsymbol{\rho})$ vanish for $\mathbf{r}$ on S may in fact not be so "simple"!)

However, situations can also arise in which even the *specification* of boundary conditions becomes difficult. Nevertheless, on the basis of physical considerations one may be able to get an idea of what to expect for $\psi(\mathbf{r})$ and its normal derivative on the boundary of some region.

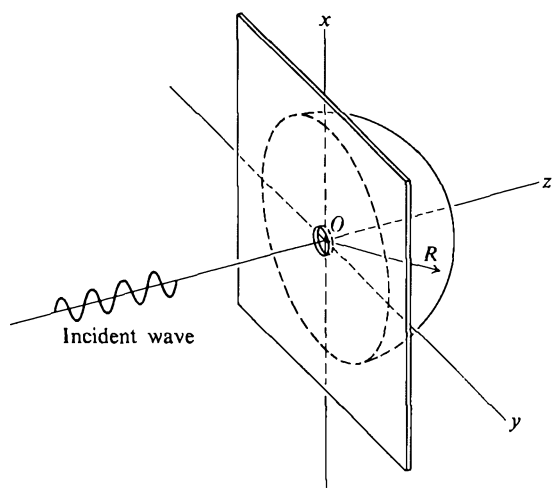

Fig. 7.8 A schematic illustration of the diffraction problem.

A beautiful example of such a situation, and one of the most important applications of Eq. (7.104), is the study of the diffraction of waves by a boundary. From a rigorous mathematical point of view, the theory of diffraction is one of the most complicated topics in classical physics. It involves solving Maxwell's equations in the presence of an awkward boundary, such as a barrier with a hole in it, and only under very special circumstances can an exact solution be obtained. However, in the freewheeling spirit of this chapter, let us try to use some physical ideas along with Eq. (7.104) to study this problem.

We envision a plane wave $e^{i\mathbf{k}_i\cdot\mathbf{r}}$ incident from the left on a flat boundary which contains an aperture, as in Fig. 7.8. The aperture is centered at the origin of the coordinate system; the screen lies in the xy-plane. Everywhere in space except on the screen the function describing the wave must satisfy the wave equation,

$$\nabla^2\psi + k_i^2\psi = 0 .$$

We are interested in the solution to the right of the screen, so let us take as our volume a hemisphere of large radius centered at the origin, whose flat bottom lies infinitesimally close to the screen.

Now what can we say about the nature of the solution on the boundary? Certainly we would expect the solution at large distances from the scattering center (the aperture) to look like an outgoing spherical wave. Thus, near the curved part of the hemisphere, the wave should have the form

$$\psi(\mathbf{r}) \sim f(\theta, \phi)\frac{e^{ik_ir}}{r} . \tag{7.105}$$

Thus the appropriate Green's function for the wave equation in that region is

$$G(\mathbf{r}, \boldsymbol{\rho}) = -\frac{1}{4\pi}\frac{e^{ik_i|\mathbf{r}-\boldsymbol{\rho}|}}{|\mathbf{r}-\boldsymbol{\rho}|} , \tag{7.106}$$

which will, according to previous arguments, yield outgoing spherical waves in Eq. (7.104). What about the flat bottom surface of the hemisphere? Here we make the physically reasonable guess that on the back of the "opaque" screen (opposite from the side on which the incident wave is falling) the solution should be identically zero. In the aperture, we take the incoming wave itself as a first approximation:

$$\psi(\mathbf{r}) = e^{i\mathbf{k}_i\cdot\mathbf{r}} .$$

We now use these physical assumptions to write Eq. (7.104) in the form appropriate to this case. We have

$$\begin{aligned}\psi(\boldsymbol{\rho}) = &\int_{ap} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS \\ &+ \int f(\theta, \phi)\left[\frac{e^{ik_ir}}{r}\frac{\partial}{\partial r}G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\frac{\partial}{\partial r}\frac{e^{ik_ir}}{r}\right]_{r=R} R^2\, d\Omega .\end{aligned} \tag{7.107}$$

In the first integral, $\mathbf{n}$ is a unit vector in the negative z-direction (an *outward* normal), and the integral ranges over the entire aperture. The second integral is an angular integration over the curved surface of the hemisphere. The derivatives in the second integral are radial derivatives, since the outward normal is in the radial direction for a spherical surface. R is the radius of the hemisphere, and may be taken to be arbitrarily large. We want to show that this second term vanishes as $R \to \infty$. It is not sufficient to say that the solution drops off like $1/r$ at large distances, since there is a factor of r^2 in the volume element. However, using the value of $G(\mathbf{r}, \boldsymbol{\rho})$ given in Eq. (7.106), it is a simple matter to show that the second integral in Eq. (7.107) falls off like $1/R$. Thus this term vanishes as we make the radius of the hemisphere large, and

$$\psi(\boldsymbol{\rho}) = \int_{\text{ap}} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS\,. \tag{7.108}$$

To simplify the work, let us assume that $\mathbf{k}_i$ is in the z-direction, which means that the wave is incident perpendicular to the screen. Now $\mathbf{n}$ is in the negative z-direction, and $\mathbf{k}_i\cdot\mathbf{r} = 0$, because the surface over which we integrate is in the xy-plane. Therefore, Eq. (7.108) reduces to

$$\psi(\boldsymbol{\rho}) = \int_{\text{ap}} \left[ik_i G(\mathbf{r}, \boldsymbol{\rho}) - \frac{\partial}{\partial z} G(\mathbf{r}, \boldsymbol{\rho})\right] dS\,. \tag{7.109}$$

Even in this form the integration represents a formidable problem, so we shall simplify life further by considering $|\boldsymbol{\rho}|$ to be large compared to a typical screen dimension. Then we can use the asymptotic form of $G(\mathbf{r}, \boldsymbol{\rho})$ discussed in Section 7.4:

$$-\frac{1}{4\pi}\frac{e^{ik_i|\mathbf{r}-\boldsymbol{\rho}|}}{|\mathbf{r}-\boldsymbol{\rho}|} \xrightarrow[|\boldsymbol{\rho}|\to\infty]{} -\frac{1}{4\pi}\frac{e^{ik_i\rho}}{\rho} e^{-i\mathbf{k}_f\cdot\mathbf{r}}\,, \tag{7.110}$$

where $\mathbf{k}_f$ is a vector whose magnitude is k_i and whose direction is given by (θ, ϕ)—the same angles specifying the orientation of $\boldsymbol{\rho}$. Using the fact that

$$(\mathbf{k}_f)_z = |\mathbf{k}_f| \cos\theta = k_i \cos\theta\,,$$

and inserting Eq. (7.110) into Eq. (7.109), we get

$$\psi(\boldsymbol{\rho}) = -\frac{ik_i}{4\pi}\frac{e^{ik_i\rho}}{\rho}(1 + \cos\theta)\int_{\text{ap}} e^{-i\mathbf{k}_f\cdot\mathbf{r}}\, dS\,. \tag{7.111}$$

This is the general expression for the diffraction of a wave by an aperture in the region whose distance from the origin is much larger than the size of the aperture. The precise form of the solution depends, of course, on the geometrical details. However, several important results follow immediately from the general result of Eq. (7.111).

First, it can be shown that if the wavelength λ of the radiation is small compared to the aperture dimensions, there is very intense radiation in the forward direction. In this direction, $\theta = 0$, and $\mathbf{k}_f\cdot\mathbf{r} = 0$, because $\mathbf{k}_f$ is per-

pendicular to the screen, on which $\theta = \pi/2$. Therefore

$$[\psi(\boldsymbol{\rho})]_{\text{forward}} = -\frac{ik_i A}{2\pi}\frac{e^{ik_i\rho}}{\rho}, \tag{7.112}$$

where A is the area of the aperture. Since $k_i = 2\pi/\lambda$, we see that

$$[\psi(\boldsymbol{\rho})]_{\text{forward}} = \psi(\rho, \theta = 0, \phi) = -i\frac{A}{\lambda}\frac{e^{ik_i\rho}}{\rho}.$$

If we denote the number of photons per square centimeter per second falling on the screen by I_0, then the number of photons per second at a distance $|\boldsymbol{\rho}| = R$ coming into the solid angle $d\Omega$ about some direction (θ, ϕ) is given by

$$I(\theta, \phi) = I_0|\psi(R, \theta, \phi)|^2 R^2 d\Omega .$$

For $\theta = 0$ this is

$$I(0, \phi) = (I_0 A)(A/\lambda^2)d\Omega ,$$

which is very large if $\lambda^2 \ll A$. Note that $I_0 A$ is just the number of photons per second falling on the aperture. If λ is much smaller than a typical aperture dimension then it is possible to evaluate the total intensity of diffracted radiation by integrating $I(\theta, \phi)$ over all solid angle, using Eq. (7.111) for $\psi(R, \theta, \phi)$. This integration yields precisely $I_0 A$, i.e., the number of diffracted photons per second arriving on the hemisphere is exactly equal to the number of photons per second falling on the screen, in the short wavelength limit.

It will also become apparent in the following examples that $\psi(\rho, \pi/2, \phi)$ —the solution on the screen—and its normal derivative are very small in absolute value compared to $|\psi(\rho, 0, \phi)|$. Thus the boundary condition on the screen is approximately, although not exactly, satisfied in this case.

Let us now see what $\psi(\boldsymbol{\rho})$ looks like for some simple aperture shapes. The easiest case is that of a rectangular aperture with sides a and b, centered about the origin. Then

$$\begin{aligned}\psi(\boldsymbol{\rho}) &= -\frac{ik_i}{4\pi}(1 + \cos\theta)\frac{e^{ik_i\rho}}{\rho}\int_{-a/2}^{a/2} e^{ik_x x}\,dx \int_{-b/2}^{b/2} e^{ik_y y}\,dy \\ &= -\frac{ik_i}{\pi}(1 + \cos\theta)\frac{\sin(k_x a/2)\sin(k_y b/2)}{k_x k_y}\frac{e^{ik_i\rho}}{\rho}.\end{aligned}$$

This has the form of Eq. (7.112) in the forward direction ($\theta = 0$) since

$$k_x = k_i \sin\theta\cos\phi$$

and

$$k_y = k_i \sin\theta\sin\phi.$$

It is also of interest to consider $\psi(\boldsymbol{\rho})$ on the screen (at $\theta = \pi/2$). We find that

$$\psi(\rho, \theta = \pi/2, \phi) = -\frac{i}{\pi k_i}\frac{\sin[(k_i a/2)\cos\phi]\sin[(k_i b/2)\sin\phi]}{\sin\phi\cos\phi}\frac{e^{ik_i\rho}}{\rho}.$$

If λ is small, k_i is large, so $\psi(\rho, \theta = \pi/2, \phi)$ is small on the screen, as we said above. The derivative normal to the screen is also very small when $\theta = \pi/2$. Thus the boundary conditions on the screen are approximately satisfied, at least for large ρ.

A particularly convenient arrangement is a circular aperture. Denoting a point in the aperture by $(r, \theta_r = \pi/2, \phi_r)$ and $\mathbf{k}_f$ by (k_i, θ, ϕ), we have

$$\psi(\boldsymbol{\rho}) = -\frac{ik_i}{4\pi}\frac{e^{ik_i\rho}}{\rho}(1 + \cos\theta)\int_0^R r\,dr\int_0^{2\pi} d\phi_r e^{-ik_i r\sin\theta\cos(\phi_r-\phi)} ,$$

where R is the radius of the aperture. Here we have used the fact that if α is the angle between two vectors whose directions are specified by (θ, ϕ) and (θ_r, ϕ_r), then

$$\cos\alpha = \cos\theta\cos\theta_r + \sin\theta\sin\theta_r\cos(\phi_r - \phi) .$$

Because of the rotational symmetry about the z-axis, the final result cannot depend on ϕ, so by changing variables to $x = k_i r\sin\theta$, we get

$$\psi(\boldsymbol{\rho}) = -\frac{i}{2k_i}\frac{e^{ik_i\rho}}{\rho}\frac{(1 + \cos\theta)}{\sin^2\theta}\int_0^{k_i R\sin\theta} x\,dx\left[\frac{1}{2\pi}\int_0^{2\pi} e^{-ix\cos\beta}\,d\beta\right] .$$

The term in brackets is easily transformed into the integral representation of the Bessel function of order zero, (Section 6.9). Thus

$$\psi(\boldsymbol{\rho}) = -\frac{i}{2k_i}\frac{(1 + \cos\theta)}{\sin^2\theta}\int_0^{k_i R\sin\theta} xJ_0(x)\,dx\,\frac{e^{ik_i\rho}}{\rho} .$$

Using Eq. (6.69), which gives an infinite series for the Bessel functions of integral order, the reader can show that

$$xJ_0(x) = \frac{d}{dx}[xJ_1(x)] ,$$

so finally

$$\psi(\boldsymbol{\rho}) = -\frac{iR}{2}\frac{(1 + \cos\theta)}{\sin\theta}J_1(k_i R\sin\theta)\frac{e^{ik_i\rho}}{\rho} . \tag{7.113}$$

This agrees with the general result of Eq. (7.112) when $\theta = 0$, for, as can be seen from Eq. (6.69), $J_1(x) \to x/2$ as $x \to 0$. Since $J_1(x) \sim 1/\sqrt{x}$ for large x, we can see that when $\theta = \pi/2$, $\psi(\boldsymbol{\rho})$ and its derivative normal to the screen are very small if the wavelength is small compared to R. Once again, the boundary conditions on the screen are approximately satisfied. The Bessel functions are oscillatory, so if we put a screen behind the circular aperture at a distance that is large compared to the radius of the aperture, the radiation falling on it will give rise to circular rings, starting with an intense central bright ring, then a dark ring as the Bessel function passes through its first zero, and so forth. Figure 7.9 shows the quantity

$$|\psi(\rho, \theta, \phi)|^2\rho^2 ,$$

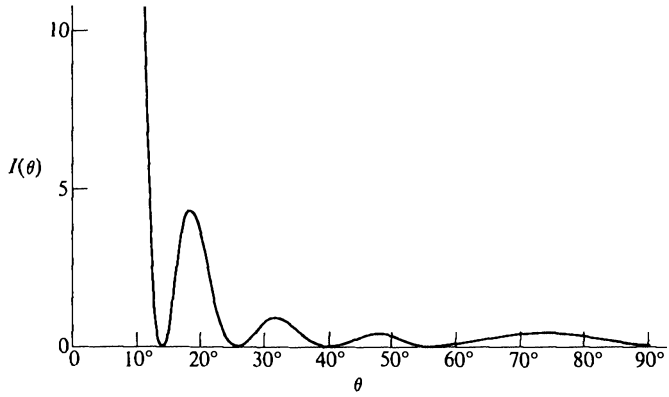

Fig. 7.9 The intensity of diffracted radiation in arbitrary units as a function of angle. The curve does not reach $\theta = 0$ until $I = 256$. This curve corresponds to a circular aperture with $kR = 16$ in Eq. (7.113). Essentially all the intensity is concentrated in the first peak.

which is proportional to the intensity; it is also independent of ϕ because of the symmetry about the z-axis. The intensity pattern of Fig. 7.9 has been verified experimentally.

It is interesting to ask the question, "What would we find if instead of having radiation fall on a screen containing a hole, we let it fall on a small screen exactly the size of the hole?" A screen of this type is called the screen *complementary* to the one we have been considering; a screen plus its complement form a completely opaque barrier. To solve this problem we use the configuration of Fig. 7.8 except that the "screen" is now transparent and the "aperture" is now opaque. In order that the boundary conditions in the xy-plane be consistent with our previous treatment, we take ψ_c to be given simply by $e^{i\mathbf{k}_i\cdot\mathbf{r}}$ in the transparent part, and identically equal to zero behind the opaque part. On the curved hemispherical surface we expect at large distances to find a scattered spherical wave superimposed on the incident plane wave:

$$\psi_c(\mathbf{r}) \sim e^{i\mathbf{k}_i\cdot\mathbf{r}} + F(\theta, \phi)e^{ik_i r}/r\,.$$

From Eq. (7.104) we have for the "complementary solution," $\psi_c(\boldsymbol{\rho})$,

$$\begin{aligned}\psi_c(\boldsymbol{\rho}) = &\int_{xy-\mathrm{ap}} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS \\ &+ \int_{\mathrm{hemisphere}} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS \\ &+ \int_{\mathrm{hemisphere}} F(\theta, \phi)\left[\frac{e^{ik_i r}}{r}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla\frac{e^{ik_i r}}{r}\right]\cdot\mathbf{n}\, dS\,, \qquad (7.114)\end{aligned}$$

where we have used our assumptions about the behavior of ψ on the boundaries. In Eq. (7.114) the subscript $xy - \mathrm{ap}$ means that the integral goes over the entire xy-plane, minus the aperture. Now, we have seen before that the last term of

Eq. (7.114) vanishes when the radius of the hemisphere becomes very large. Thus

$$\psi_c(\boldsymbol{\rho}) = \int_{xy-\text{ap}} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS$$
$$+ \int_{\text{hemisphere}} [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS\,. \tag{7.115}$$

If we now compare this result with Eq. (7.108) for $\psi(\boldsymbol{\rho})$, we see that the sum $\psi(\boldsymbol{\rho}) + \psi_c(\boldsymbol{\rho})$ obeys the simple relation

$$\psi(\boldsymbol{\rho}) + \psi_c(\boldsymbol{\rho}) = \int_S [e^{i\mathbf{k}_i\cdot\mathbf{r}}\nabla G(\mathbf{r}, \boldsymbol{\rho}) - G(\mathbf{r}, \boldsymbol{\rho})\nabla e^{i\mathbf{k}_i\cdot\mathbf{r}}]\cdot\mathbf{n}\, dS\,, \tag{7.116}$$

where S denotes, as before, the *entire* closed surface. Now using Green's theorem and remembering that $e^{i\mathbf{k}_i\cdot\mathbf{r}}$ is a solution of the wave equation, we see that

$$\psi(\boldsymbol{\rho}) + \psi_c(\boldsymbol{\rho}) = e^{i\mathbf{k}_i\cdot\boldsymbol{\rho}}\,. \tag{7.117}$$

This beautifully simple result is known in optics as *Babinet's principle* or *the principle of complementary screens.*

If we use Eq. (7.111) for $\psi(\boldsymbol{\rho})$ at distances that are large compared to the dimensions of the screen, then

$$\psi_c(\boldsymbol{\rho}) = e^{i\mathbf{k}_i\cdot\boldsymbol{\rho}} + f(\theta, \phi)\,\frac{e^{ik_i\rho}}{\rho}\,, \tag{7.118a}$$

where

$$f(\theta, \phi) = \frac{ik_i}{4\pi}(1 + \cos\theta)\int_{\text{ap}} e^{-i\mathbf{k}_f\cdot\mathbf{r}}\, dS\,. \tag{7.118b}$$

Equations (7.118a) and (7.118b) are of the same form as the results which we found in our study of the Lippmann-Schwinger equation. In the latter case, $f(\theta, \phi)$, the scattering amplitude, was given in terms of the potential and the wave function. Here we obtain $f(\theta, \phi)$ in terms of a boundary condition.

According to Eq. (7.113), the scattering of a plane wave by a circular, opaque obstacle is described by the function

$$f(\theta, \phi) = \frac{iR}{2}\,\frac{1 + \cos\theta}{\sin\theta}\, J_1(k_i R \sin\theta)\,, \tag{7.119}$$

where R is the radius of the obstacle and $k_i = 2\pi/\lambda$, as before. Several interesting attempts have been made to understand in terms of diffraction the quantum-mechanical scattering of short wavelength (high energy) nuclear projectiles (α-particles, for example) from nuclei. For reasonable values of the nuclear radius, scattering amplitudes given by Eq. (7.119) match the experimental results quite closely. They show a strong forward maximum and an oscillatory behavior away from the forward peak very similar to the pattern shown in Fig. 7-8. This provides a nice illustration of the wave aspect of particles.

7.7 TIME-DEPENDENT GREEN'S FUNCTIONS: FIRST ORDER

In the past few sections we have concentrated our attention of the application of Green's function techniques to the solution of certain partial differential equations in which the time did not play an essential role. In this section we begin looking at problems where the evolution of some system in time (as well as in space) is important.

Consider an equation such as

$$H\psi + \frac{\partial \psi}{\partial \tau} = 0 , \tag{7.120}$$

where τ is a time variable, and H is an operator, *independent of time* which is assumed to possess a complete set of eigenfunctions. There are numerous equations of this type, and their application extends into all domains of physics. A familiar example is the heat equation,

$$\nabla^2 T = \frac{c}{k}\frac{\partial T}{\partial t} ,$$

where T is the temperature, c is the specific heat per unit volume, and k is the heat conductivity. If we call $H = -\nabla^2$ and $\tau = kt/c$, then this has precisely the form of Eq. (7.120). Schrödinger's time-dependent equation is also of this form, namely,

$$\mathscr{H}\psi = i\hbar \frac{\partial \psi}{\partial t} ,$$

where $\mathscr{H}$ is the total Hamiltonian of the system described by the wave function ψ. For example, for a free particle,

$$\mathscr{H} = -\frac{\hbar^2}{2m}\nabla^2 ,$$

so Schrödinger's equation is

$$-\nabla^2\psi = \frac{2mi}{\hbar}\frac{\partial \psi}{\partial t} .$$

Thus if we set $H = -\nabla^2$ and $\tau = (i\hbar/2m)t$, the equation has exactly the same form as the heat equation. Another equation of this type is the diffusion equation which reads

$$\nabla^2 \rho = \frac{1}{a^2}\frac{\partial \rho}{\partial t} ,$$

where ρ is the density of the diffusing material, and a is the diffusion constant. Here again the identification $H = -\nabla^2$ and $\tau = a^2 t$ brings the equation to the basic form of Eq. (7.120).

Since we assume that H has a complete orthonormal set of eigenfunctions,

$$H\phi_m = \lambda_m \phi_m ,$$